

SENIOR Geography 1

AUSTRALIA

Kate Lanceley & Bruce Pacey



Quality ready-to-use
resources

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INTRODUCTION

Instant Lessons in Senior Geography: Book 1 – Australia is part of a two-book series, which is a collection of student friendly worksheets created for Years 9–12 Geography students throughout Australia.

The worksheets explore the physical, environmental, social and political geography of Australia. They are to be used in conjunction with set texts and classroom studies. They can be used in a variety of ways, either to introduce a topic, as a review of issues already studied, to further expand students’ knowledge and skills, a summary or as a replacement lesson when a teacher is away.

Emphasis has been placed on skills development. Students are required to interpret stimulus material using geographical tools. A variety of different learning activities are included in each worksheet to engage students’ interest.

Some worksheets provide space for student answers, but often a separate sheet or workbook is required. A blank outline map of Australia is provided on the inside front cover to be used in topics where this is noted. A map of Asia is also provided on the inside back cover for student use.

CONTENTS

1	Introducing Australia’s physical geography	4
2	Introducing Australia’s human geography	6
3	Landform	8
4	Water resources and drainage	10
5	Air masses and climate	12
6	Weather patterns	14
7	Vegetation and forest resources	16
8	Soils and salinity	18
9	Natural hazards	20
10	Drought and bushfires	22
11	Demographic characteristics	24
12	Population distribution	26
13	Indigenous Australians	28
14	Native title	30
15	Australia’s changing age structure	32
16	Industrial landscape	34
17	Regional and global links – defence	36
18	Regional and global links – international tourism	38
19	Regional and global links – migration	40
20	Regional and global links – trade	42
	Selected answers	44

1 INTRODUCING AUSTRALIA'S PHYSICAL GEOGRAPHY

Note to teacher

Students will need to use an atlas and a blank outline map of Australia to complete this worksheet.

Locating Australia

1a Fill in the blanks.

Australia is located between latitudes

(1) _____ S and (2) _____ S and longitudes

(3) _____ E and (4) _____ E.

The Australian continent extends

(5) _____ kilometres east to west and

(6) _____ kilometres north to south.

Unique characteristics of Australia

2a Fill in the blanks below. Australia is:

∞ the world's (1) _____ island with a land area of
(2) _____ km²

∞ the world's (3) _____ continent. Its highest point is
(4) _____ metres.

∞ the world's second (5) _____ continent after
Antarctica.

2b Use an atlas to name the following:

Australia's:

∞ most northern point (1) _____

∞ most southern point (2) _____

∞ most eastern point (3) _____

∞ most western point (4) _____

∞ highest mountain (5) _____

∞ longest river (6) _____

Australia's climate

Climate varies enormously throughout Australia. These variations can be broadly attributed to:

∞ latitude

∞ altitude

∞ distance from the coast.

3 Suggest possible reasons for the occurrences listed in
a–e in the right column:

a Darwin has warm temperatures all year round with a distinct wet season from December to March and a distinct dry season from May to August.

b The temperature drops as you climb Mount Kosciuszko.

c Alice Springs has low rainfall all year.

d Sydney has warmer temperatures on average than Hobart.

e Perth has hot dry Summers.

Australian landforms

4a Australia’s landforms can be broadly divided into three areas:

The (1) _____ Plateau

The (2) _____ Plains and the

(3) _____ Highlands

Within these areas there are many specific landform features.

4b Match the following landform examples with the correct type of landform feature.

Carnarvon Gorge, Devils Marbles, Eastern Highlands, Flinders Ranges, Kimberley, Mount Gambier, Nullarbor, Uluru

Landform features	
fold mountains	
block mountains	
volcano	
dissected plateau	
granite boulders	
limestone plain	
sandstone monolith	
sedimentary plateau	



Who am I?

5 Name the places in a–h at right and mark each one on a blank outline map of Australia.

a A 260-metre long reef. I am the largest fringing coral reef in Australia. I contain over 500 species of fish and 200 species of coral. _____

b Located 110 km south west of Adelaide. I am Australia’s third largest island. My natural attractions make me very popular with tourists. _____

c I am the world’s largest sand island. I am 125 km long and over 160 000 hectares in area. I was formed during the ice age when the prevailing winds transported vast quantities of sand from New South Wales and deposited it along the coast of Queensland. _____

d I run 300 km, east and west of Alice Springs. I am the second longest and highest mountain range in Australia characterised by red sandstone gorges and waterholes lined with red river gums and ghost gums. _____

e Believed to have formed 2700 million years ago. I am 14 metres high and 110 metres long and I look like I am just about to break. _____

f Located 987 km West of Sydney. I am one of 17 dry lakes which constitute the Willandra Lakes World Heritage Area. I dried up 14 000 years ago and am now a rich source of fossils. _____

g I am a 135-million-year-old rainforest located in Tropical Far North Queensland. I was placed on the World Heritage list in 1988. _____

h I am 1545 metres tall and one of the most popular tourist destinations in Tasmania. _____

True or false?

6 Label the following statements true or false. Rewrite the incorrect statements correctly.

Statement	T or F?	Correct statement
a Cape York Peninsula is north-west of Alice Springs.		
b Melville Island is Australia’s second largest island after Tasmania.		
c The Flinders Ranges are located in Western Australia.		
d Mt Kosciuszko is Australia’s highest point.		
e It is much cooler in the northern parts of Australia than in the south.		
f Perth’s climate can be described as tropical and wet.		

2

INTRODUCING AUSTRALIA'S HUMAN GEOGRAPHY

Note to teacher

Students will need to use an atlas and a blank outline map of Australia to complete this worksheet.

Locating cities and towns

1a Use an atlas to name the cities and towns found at the following locations.

Geographic coordinates	Place	Geographic coordinates	Place
23°22'S 150°32 E	1	34.86S 138.36E	6
31°57'S 115°51'E	2	21.09S 149.11E	7
23°42'S 133°52'E	3	42.50S 147.15E	8
20°44'S 139°29'E	4	31.58S 141.27E	9
38°10'S 144°21'E	5	35.00S 117.52E	10

1b Match each of the following cities/towns with the state in which it is found. Also, in column 2, write the accepted abbreviation for each Australian state and territory.

Albany, Alice Springs, Bairnsdale, Belconnen, Broken Hill, Broome, Cairns, Coober Pedy, Coonabarabran, Darwin, Devonport, Kalgoorlie, Katherine, Launceston, Longreach, Mount Gambier, Portland, Queenstown, Shepparton, Tamworth, Tuggeranong, Weipa, Weston Creek, Whyalla
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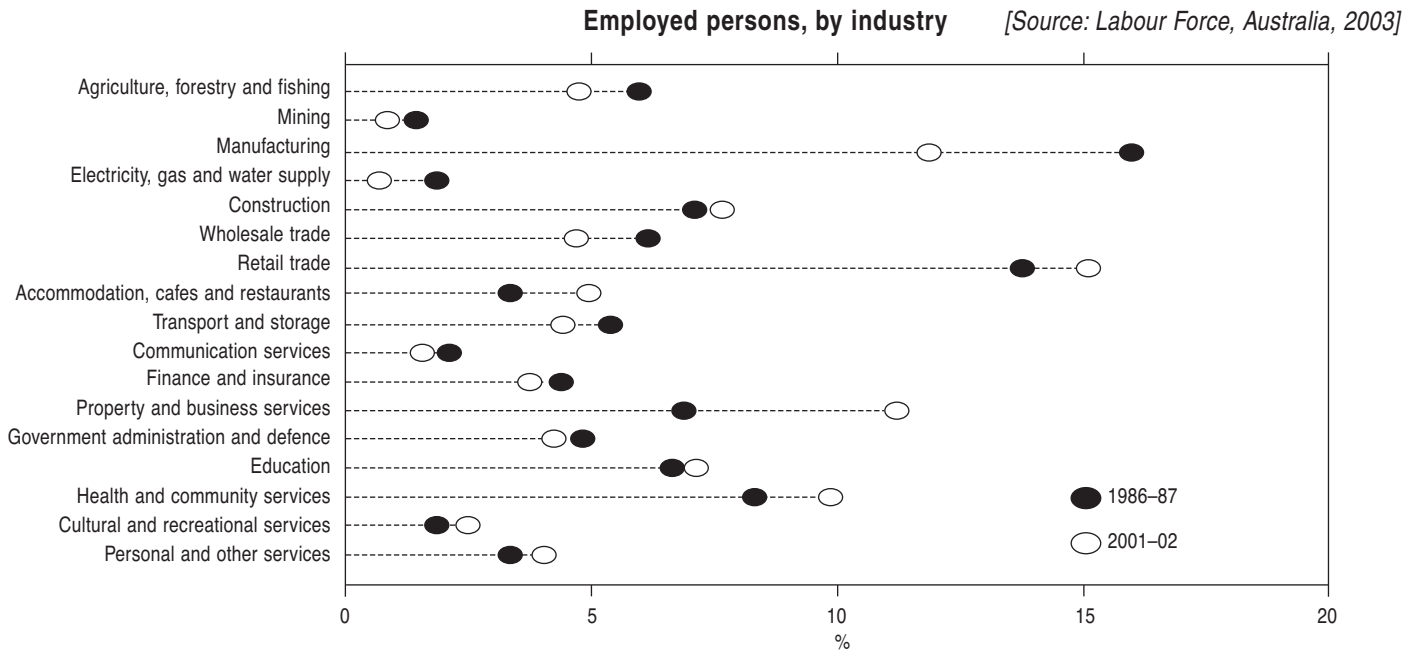
State or territory	Abbr.	City/town	City/town	City/town
ACT		1	9	17
New South Wales		2	10	18
Northern Territory		3	11	19
Queensland		4	12	20
South Australia		5	13	21
Tasmania		6	14	22
Victoria		7	15	23
Western Australia		8	16	24

Australia's population

2 Match each of the following cities with its approximate population (for 2004): In column 3 write a short description of the location of each city using attributes more than just geographic coordinates: Adelaide, Brisbane, Canberra, Darwin, Hobart, Melbourne, Perth, Sydney

City	2004 population ('000)	Location
	4 085.6	
	3 466.0	
	1 626.9	
	1 381.1	
	1 096.1	
	310.5	
	194.2	
	90.0	

Industry



3 Use the graph above to identify the 3 largest industries in Australia by number of people employed in

a 1986-87

b 2001-02

Who am I?

4a I am one of Australia's oldest industries. While once Australia was 'riding on its back', in recent years the industry has experienced decline. Australia, is still however the world's largest producer, accounting for over a quarter of world production.

4b I am the second largest settlement in NSW after Sydney. I am a river port which specialises in bulk cargo, with my biggest export being coal.

4c I am an isolated mining town, located 838 kilometres north of Cairns. I was built for the purpose of mining bauxite, which is mined, washed, graded and loaded here for shipment to an aluminium processing plant at Gladstone.

4d Due to globalisation and trade liberalisation, I have experienced substantial decline in recent decades as I am no longer able to compete globally.

4e I have experienced growth in recent decades and am a substantial contributor to Australia's exports and gross domestic product. I was given a boost by Australia's hosting of the Olympics and the Rugby World Cup.

4f This industry has experienced spectacular growth in recent years. Australia now ranks in the top six producers in the world. Our top three producing zones are the Lower Murray (SA), Big Rivers (NSW) and North West Victoria.

Multiple choice

5 Australia's largest state/territory in land area is:

- A** New South Wales **B** Northern Territory
C Queensland **D** Western Australia

6 Australia's largest state/territory in population is:

- A** New South Wales **B** Queensland
C Western Australia **D** Victoria

7 Which Australian city has the largest share of its states population?

- A** Adelaide **B** Brisbane
C Melbourne **D** Sydney

8 The industry to experience the largest decline in employment in Australia between 1986-87 and 2001-02 was

- A** mining
B manufacturing
C property and business services
D cultural and recreational services

9 The industry to experience the largest increase in employment in Australia between 1986-87 and 2001-02 was

- A** manufacturing
B retail trade
C property and business services
D health and community services

3

LANDFORM

Introduction Landform refers to the shape of the earth’s surface, and is the result of processes operating both beneath the earth surface (tectonic forces) and those affecting the land from above (gradational forces). Tectonic forces usually force the surface upwards or change its structure, while gradation forces usually break it down and even out the level of the land. Individual landform features are often classified into residual, erosional or depositional by the way these gradational forces have affected the initial surface.

Processes shaping landforms

1 Classify the following landform processes into either tectonic or gradation forces.

Process	Type: tectonic or gradation
weathering	
folding	
faulting	
erosion	
vulcanism	
landslide	
deposition	

2 Residual features are those remaining after, or resisting gradation. Erosional features are created where material has been removed, while depositional features consist of material relocated following its erosion. In the table below match the following landforms with their description, and indicate if they are residual, erosional or depositional.

alluvial fan, beach, canyon, cirque, crater, delta, dune, gibber plain, mesa, mountain, plain, plateau, ridge, valley

	Landform	Description	Type
a		an area of lower land, usually between hills or mountains	
b		an accumulation of material at a river-mouth	
c		a flat-topped hill	
d		a deep, steep sided valley	
e		a stony surface where wind has removed finer materials	
f		an elongated and elevated section of land	
g		an accumulation of material found spreading where a stream emerges from highland onto a plain	
h		a large, flat, low-lying area	
i		a depression in a mountainside caused by glacial activity	
j		material that accumulates along a coastline	
k		a very high, steeply sloped landform feature	
l		an undulating pattern of windblown material	
m		a large, flat, elevated area	
n		the depression found at a volcanic peak	

Elevation

Australia is the lowest and flattest of all the continents. This is because it is located almost centrally on a crustal plate, with no major mountain-building activity that commonly occurs where plates join. The ancient rocks have therefore been subjected to gradational forces for much longer than on other continents.

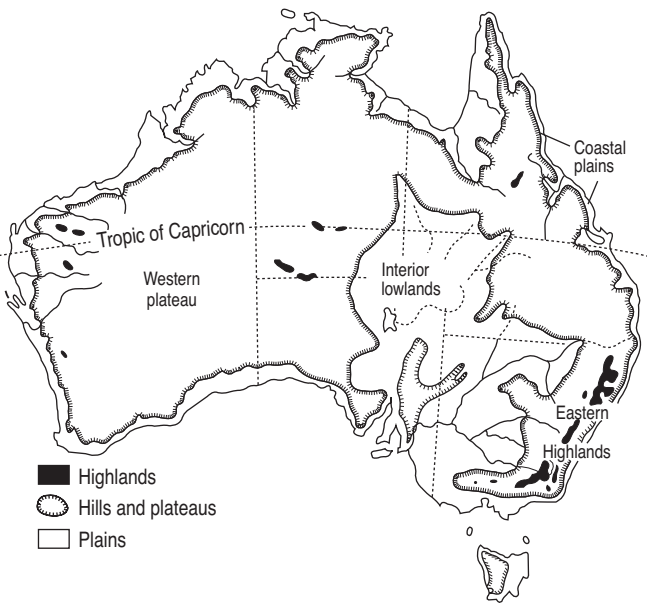
3a Using the data provided below construct a column graph to show the height of each of the continents.

3b Calculate a world average from each of the continental figures. What is your result ? Explain why it is different to the world average in the table.

Continent	Average elevation (metres)
Australia	300
Europe	340
South America	590
North America	720
Africa	750
Asia	960
Antarctica	2200
WORLD AVERAGE	700

Multiple choice

Australia’s landform regions



Refer to the map above. Circle the letter to the answer that best suits each of the questions 4–8 in the right column.

- 4 The type of landform covering the largest area of Australia is:
A coastal plains
B hills and plateaus
C interior lowlands
D highlands
- 5 In a journey across Australia from west to east, following the Tropic of Capricorn, the main types of landform you would cross, according to the map, in order are:
A coastal plains, highlands, interior lowlands, hills and plateau, coastal plains
B coastal plains, interior lowlands, hills and plateaus, highlands
C coastal plains, hills and plateau, interior lowlands hills and plateau, coastal plains
D coastal plains, interior lowlands, highlands coastal plains
- 6 Australia’s coastal plains are widest in which of the following regions?
A The Gulf of Carpentaria
B The Great Australian Bight
C The SW coast of Western Australia
D The NE coast of Queensland
- 7 Desert landforms, such as sand dunes, mesas, gibbers and canyons would most likely be located in the:
A highlands and interior lowlands
B interior lowlands and coastal plains
C interior lowlands and western plateau
D western plateau and highlands
- 8 Australia has ancient, extinct volcanoes in the eastern states which are oldest in Queensland, of intermediate age in New South Wales, and youngest in Victoria and South Australia. This most likely provided evidence that Australia:
A is drifting closer to New Zealand which has active volcanoes
B has been drifting northward over a “hot-spot” in the earth’s crust
C has been drifting southward over a “hot-spot” in the earths crust
D is about to experience renewed volcanic eruptions

Extended response

- 9 Using the points below as a guide for sections or paragraphs, write an extended response explaining how landforms can influence other features of geography. Include as many examples from your knowledge as you can:
 - Landuse
 - Transport
 - Settlement

4

WATER RESOURCES AND DRAINAGE

Introduction Australia’s low rainfall and the distribution of water resources are the most important factors determining our settlement pattern, landuse and rural production. A scarcity of both surface and groundwater resources places significant limitations on our use of this continent that can only partially be overcome with modifications such as dams and reservoirs. Although our main river system, the Murray–Darling, ranks as the world’s 15th longest, its discharge (the volume of water flowing through it) is among the lowest for the world’s major river systems. Recent and current water shortages and restrictions on water use in Australian towns and cities have highlighted our water resource problems.

Terminology

1 Match each of the following terms with the best definition in the table.

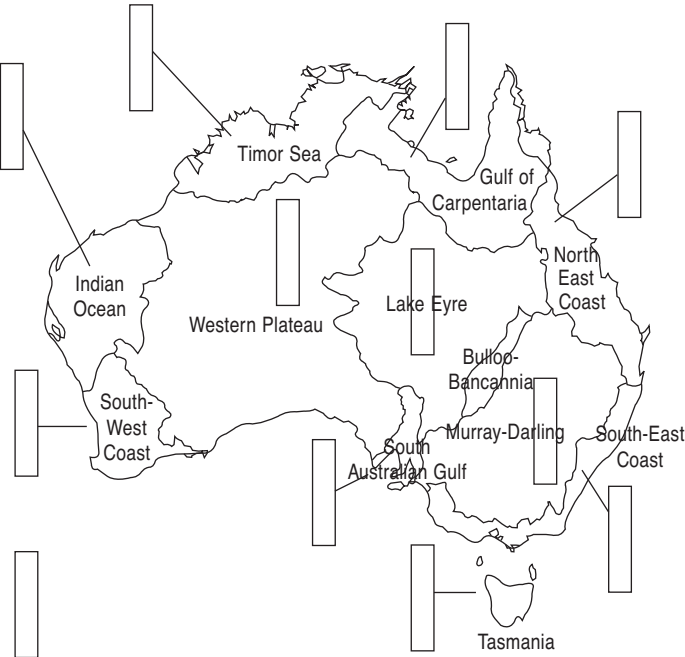
aquifer, artesian, catchment, discharge, external, gigalitre, intermittent, internal, irrigation, perennial

Term	Definition	Term	Definition
	1 an addition of water to the naturally available supply of an area, usually for growing of crops		6 water that travels through an aquifer to be released elsewhere in the landscape either naturally or by drilling a bore
	2 a rock layer which absorbs and transports underground water		7 describes rivers or river systems which do not flow to the ocean
	3 water volume equal to one thousand million litres		8 the area of land from which a river or river system gains its water
	4 the term for rivers which are constantly flowing		9 the amount of water flowing through or out of a river
	5 rivers which drain to an outlet at the sea		10 the term for rivers which cease to flow for a period of time, such as a dry season

Water use

Australia’s drainage regions

[Adapted from the Australian Natural Resources Atlas Australian Government – 2004]



Drainage Region	Total annual surface water use – gigalitres 1996–97	Urban and industrial use %	Irrigation, rural use %
North-east coast	2144	35	65
South-east coast	1784	74	26
Tasmania	451	40	60
Murray-Darling	11 149	5	95
South Australian Gulf	272	92	8
South-west coast	362	49	51
Indian Ocean	12	83	17
Timor Sea	318	14	86
Gulf of Carpentaria	98	31	69
Lake Eyre	7	14	86
Western Plateau	14	93	7

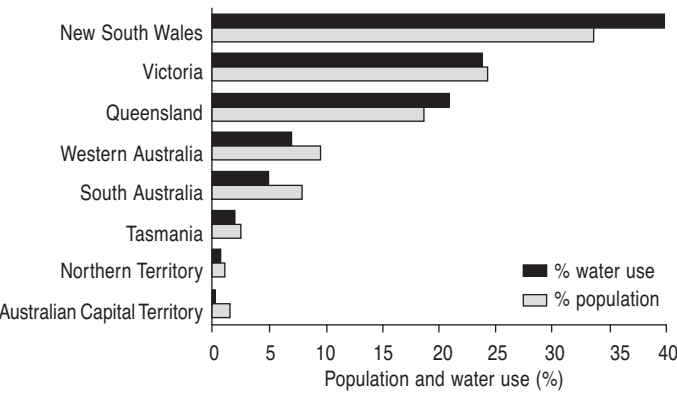
2 Using information from the table opposite, label the map above with the total water use and graph in each bar the proportions of urban/industrial and irrigation use. Colour-code the graphs and legend.

[Source: State of the Environment Report, Australian Government 2001]

Water sources

3 Groundwater is an important source in the drier regions of Australia. Construct a sector graph to show the source of total water use in Australia using the following data: Rivers 73%, Groundwater 21%, Other surface water (eg, Lakes) 6%. [Source: Australian Water resources Assessment, Australian Government 2000].

State variations in water use and resources



[Source: Australian Water Resources Assessment – Australian Government 2000]

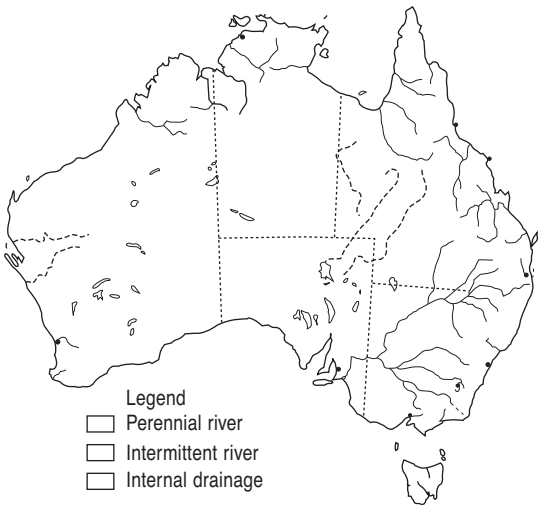
- 4a Use the graph above to briefly describe the general relationship between water use and population in Australia.
- 4b Identify the states/territories that have a higher share of water use than their share of population.
- 4c Which state or territory appears to be the most efficient user of water per head of population?
- 4d Write a paragraph response, suggesting reasons for the patterns shown in the graph. Consideration of rainfall, settlement and landuse patterns may help you with your response.

Australia’s mean annual water use (GL) by primary water source (1996–7)

	Total use surface water (GL)	Total use ground-water (GL)	% sourced from ground water
New South Wales	9 000	1 008	
Victoria	5 166	622	
Queensland	2 969	1 622	
Western Australia	658	1 138	
South Australia	746	419	
Tasmania	451	20	
Northern Territory	51	128	
Australian Capital Territory	68	5	
Total	19 109	4 962	

[Modified from the Australian Water Resource Assessment, Australian Government 2000]

- 4e Calculate and insert in the table the percentage of water that each state/territory and the nation in total sources from groundwater.
- 4f Suggest reasons for the variations in the pattern of state/territory water sources.



- 5 Complete the legend on the map to show which rivers you would expect to be perennial and intermittent. Also, colour or highlight the map and legend to show the rivers with internal drainage.

Multiple choice

- 6 The drainage region of Australia with the highest proportion of water used for irrigation is:
 - A North-east coast
 - B Murray-Darling
 - C Timor Sea
 - D Lake Eyre
- 7 The Western Plateau drainage region has
 - A mainly intermittent rivers and external drainage
 - B perennial rivers and internal drainage
 - C short perennial external drainage
 - D few or no major rivers
- 8 The state or territory which most depends on groundwater is
 - A Northern Territory
 - B Western Australia
 - C South Australia
 - D Queensland
- 9 The state or territory that uses the most water is
 - A Queensland
 - B New South Wales
 - C Victoria
 - D South Australia
- 10 Australia’s largest river system, the Murray–Darling is significant on a world scale in terms of its
 - A length and discharge
 - B discharge and catchment size
 - C length and catchment size
 - D all of these

5 AIR MASSES AND CLIMATE

Introduction Air masses are large bodies of air with relatively uniform horizontal qualities of temperature and moisture, which they acquire from that part of the earth’s surface over which they form (their source region). Air masses are moved about by winds, and interact with pressure systems and the topography of the land to bring changes in weather when a different air mass arrives. Dominating air masses also create typical seasonal weather in some parts of Australia.

Air masses influencing Australia

1a Use the data provided to complete the table at the bottom of the page.

Names

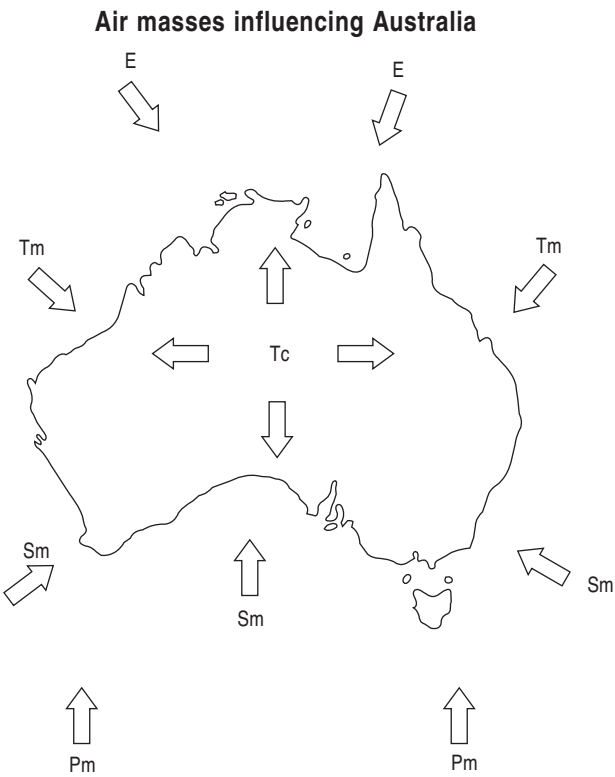
southern maritime, polar continental, equatorial, polar maritime, tropical continental, tropical maritime

Features – warm and dry, cold and dry, hot and humid, cool and humid, warm and humid, cold and humid

Region/season

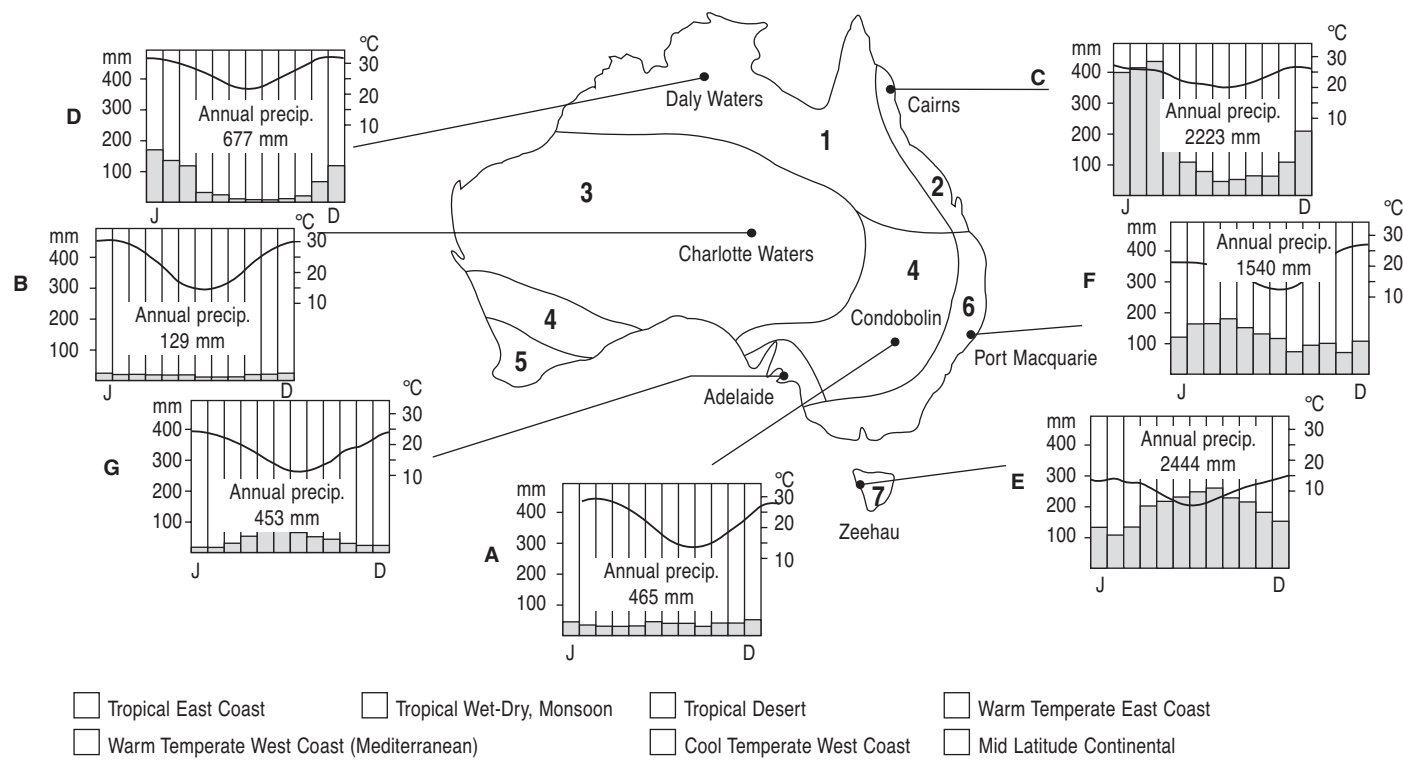
- ∞ southern areas most of the year, extending further north in winter
- ∞ northern areas in winter and further south in summer
- ∞ northern areas most of the year
- ∞ does not influence Australia
- ∞ central Australia all year
- ∞ southern areas mainly in winter

1b Develop an appropriate colour scheme for each air mass. Create a legend for the map then colour each arrow correctly.



Symbol	Name	Features	Region/season
E			
Tm			
Tc			
Sm			
Pm			
Pc			

Climatic regions



2 In the table below match the graphs and climatic region names with the areas marked on the map and label the legend boxes with the correct map number.

Climatic regions

- ∞ Mid Latitude Continental

∞ Tropical Wet/Dry (Monsoon)
- ∞ Warm Temperate West Coast (Mediterranean)

∞ Tropical Desert
- ∞ Warm Temperate East Coast

∞ Tropical East Coast

∞ Cool Temperate West Coast

Region	Graph	Climatic region
1		
2		
3		
4		
5		
6		
7		

3 Divide into 7 groups and allocate one climatic region to each group. Using the data provided in the climate graphs, the maps, describe and account for the climate of your allocated region. Include a section on what it would be like to live in each region, including seasonal changes, advantages and problems of living there, and when you think would be the best time for tourists to visit.

When each group has finished their task, share the results with the rest of the class either through an oral or computer presentation.

6 WEATHER PATTERNS

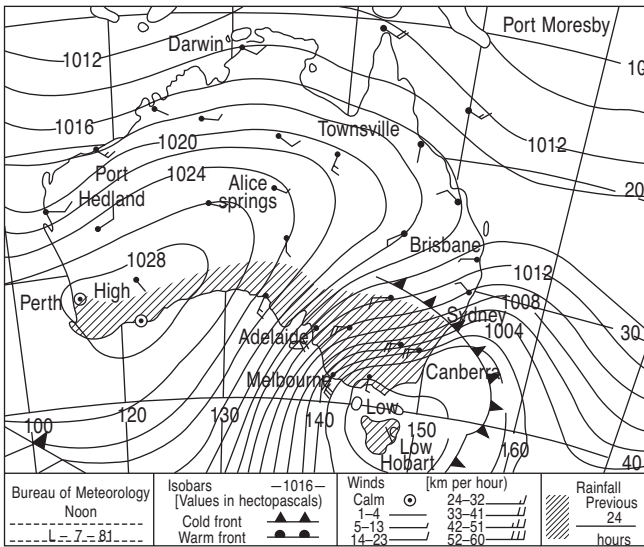
Introduction Weather refers to the short-term, day-to-day changes in the condition of the atmosphere, while climate involves the longer-term or annual patterns of weather. Many aspects of our daily lives and the operation of our society and industries are influenced by the weather. Pressure systems are a basic determinant of weather. In lows, the air rises and cools tending to create cloud, while in highs it descends and warms, bringing fine weather. Between the cells of pressure, surface winds move air masses from one place to another, creating changes in the weather. The pressure systems (and air masses) tend to migrate north and south with the seasons, creating distinct seasonal changes in the weather, known as climate.

Atmospheric conditions

- 1 Make a list of the 'conditions of the atmosphere' – that is, features of the atmosphere that can change from time to time.
- 2 Brainstorm ideas on how people and their activities are affected by weather and climate, then hold a class discussion and list the ideas on the board.

Typical winter conditions

Typical winter weather map



[Source: Bureau of Meterology]

Multiple choice

Circle the answer that best suits questions 4–7.

- 3 The air pressure in Brisbane would be closest to
 - A 1014 hPa
 - B 1013 hPa
 - C 1012 hPa
 - D 1014 isobars

- 4 The speed and direction of wind in Canberra would be
 - A 14–23 kph from the south-west
 - B 33–41 kph from the east
 - C 33–41 kph from the west
 - D 14–23 kph from the north-east
- 5 The synoptic chart feature located east of Sydney in the Tasman sea is
 - A a ridge of high pressure
 - B a warm front
 - C a cold front
 - D a ridge of low pressure
- 6 Two cities that would have experienced precipitation in the last 24 hours are
 - A Adelaide and Sydney
 - B Melbourne and Townsville
 - C Perth and Port Hedland
 - D Alice Springs and Brisbane
- 7 In the last 24 hours, Sydney would have experienced
 - A a change of wind direction
 - B a fall in temperature
 - C the onset of rainfall
 - D all of the above

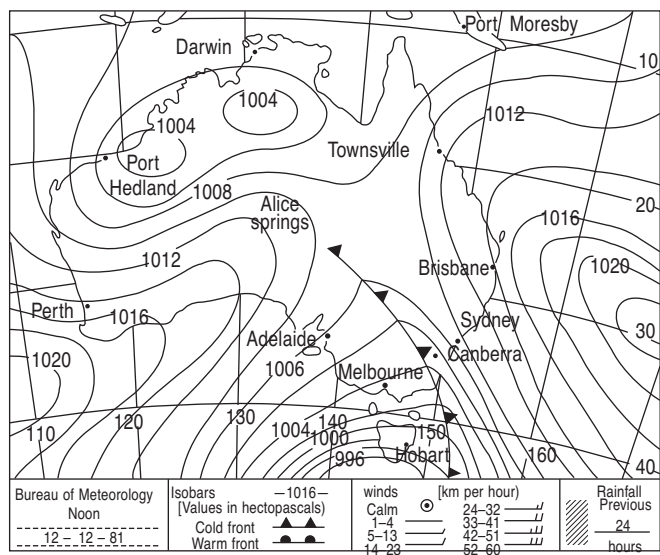
Paragraph responses

- 8 On a separate sheet compare and contrast the weather conditions being experienced in Adelaide and Darwin. Include the influence of the season as well as the current weather patterns. Consider such features as temperature, precipitation, cloud cover, wind speed and direction, and humidity. Account for differences in the weather of the two cities.
- 9 Describe how you would expect the weather in Adelaide to change over the next few days. Give reasons for your predictions.

Typical summer conditions

10 Before beginning the following questions, identify the areas of high and low pressure on the synoptic chart below and label them clearly with an H or L. Features such as rainfall, winds and pressure system labels have been removed, but can be estimated using the isobars shown.

Typical summer weather map



[Source: Bureau of Meterology]

Multiple choice

Circle the letter for the answer that best suits questions 11 to 15.

11 The synoptic feature that best indicates this is a summer chart is

- A the cold front in the south east
- B the pressure systems in the north
- C the pressure systems in the south
- D the absence of rain

12 The pressure systems located south-west of Perth and south of Tasmania are, in consecutive order

- A a cyclone and a high
- B an anticyclone and a high
- C an anticyclone and a low
- D a tropical cyclone and a low

13 Which of the following cities would be experiencing the strongest winds

- A Darwin
- B Brisbane
- C Sydney
- D Adelaide

14 The Wind direction in Perth would most likely be a

- A northerly
- B southerly
- C easterly
- D westerly

15 A state capital that would be being influenced by a SM (southern maritime) air mass would be

- A Brisbane
- B Sydney
- C Adelaide
- D Hobart

Short answer

Answer these in the spaces below or on a separate sheet of paper.

16 Describe and account for the weather you would expect Townsville to be experiencing. Use features on the synoptic chart, the season, and the location of Townsville in relation to Australia's topographic (landform) features to write as detailed a response as possible.

17 Describe and account for the weather being experienced in Sydney. How would you expect the weather to change in the next day or two. Give reasons for your predictions.

7 VEGETATION AND FOREST RESOURCES

Introduction Australia's pattern of vegetation is the result of its geologic history, past and present climate, location and size, and the impacts of its inhabitants. Land clearing and vegetation modification have accelerated to the extent that we now need replanting, pest eradication and forest protection programs. These programs are an acknowledgment of the value of our flora to our environment, economy and society.

The vegetation timeline

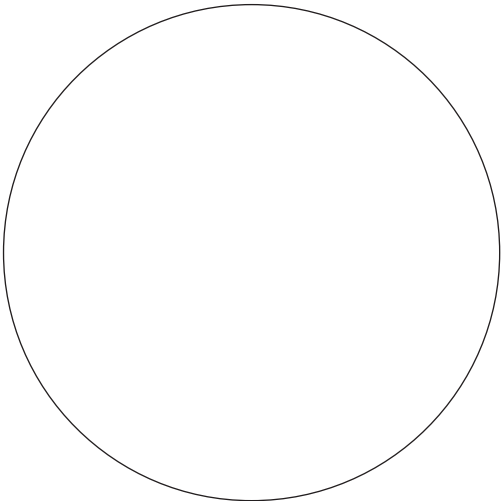
1 Place these events below in the sequence of their impact on Australia's vegetation.

Order	Event
	A Europeans settle the continent, begin clearing and introducing exotic species
	B Australia separates from Gondwana, creating a biodiversity that is about 80% endemic (not found elsewhere)
	C Following the Rio Earth Summit, Australia begins a program of Regional Forest Agreements to protect our forests and communities economically dependent on them
	D Australia drifts north into warmer zones and the climate becomes drier at the end of the last ice age
	E Aborigines enter the continent, and practise fire-stick farming, which encourages fire resistant species

Biodiversity

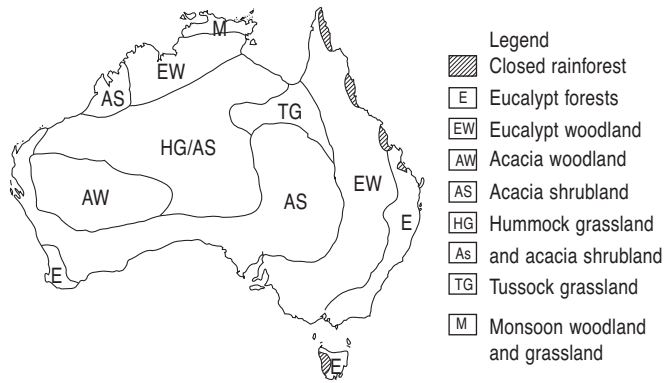
Australia's biodiversity (the range of plant species found) is relatively high with a world ranking of 17. The native genus (plant groups) with the largest number of species are: Acacia 1000, Eucalypt 850, Grevillia 260, and Melaleuca 176. The region with greatest species diversity is the rainforest.

2 Use the table at top right to show the area of the Australian continent covered by different plant communities in a sector (pie) graph.



Vegetation type	Area %
Closed forest (rainforest)	0.4
Open forest (mainly eucalypt)	4.6
Woodland (eucalypt and acacia)	39.7
Shrubland (mallee, mulga, alpine)	37.3
Grassland	9.8
Pastures (native and introduced)	6.4
Coastal communities (mangroves etc)	0.3
Other (urban, crops, no vegetation)	1.5

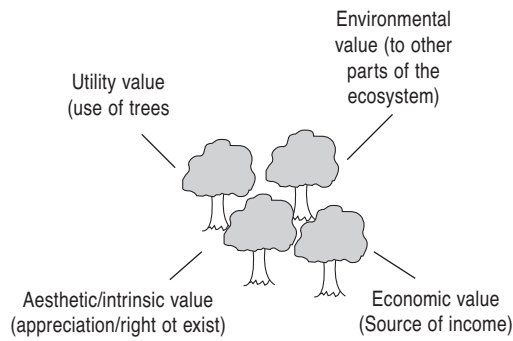
[Source: Australian National Botanic Gardens]



3 Using the map above, answer true or false to the following statements.

Statement	T or F?
Eucalypt forests are found only on the eastern side of the continent.	
Rainforests are found only on the eastern side of the continent.	
The smallest area is occupied by rainforests.	
The vegetation zones are influenced more by temperatures than by precipitation levels.	
The state with the fewest vegetation zones is Tasmania.	

Value of forests



4 Make a copy of the diagram above and expand it with examples of the value of trees.

Native forests

Native forests cover just over 5% of the Australian landmass. These have been reduced from approximately 69 to 43 million hectares since European settlement. The area of woodland has also fallen from about 157 to 112 million hectares.

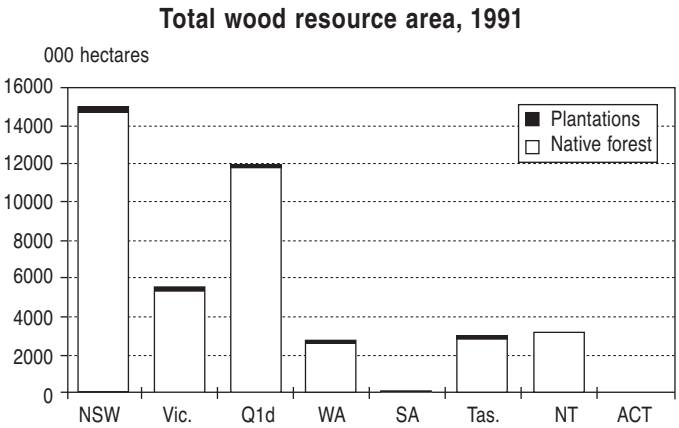
5a From the statement above calculate the proportions by which the areas of forest and woodland have been reduced. Suggest reasons for the differences in your results.

To calculate proportional or percentage change

Divide the amount of change by the original figure, and multiply by 100%.

Type of ownership	% area	Notes
Private ownership	27	small proportion allocated to logging
Public ownership		
State forests	30	multiple use, including logging reserves
Other crown land	28	logging permitted, but land not specifically allocated for logging
Conservation reserves	15	eg, National Parks – logging not permitted – these areas increasing in their share.

[Adapted from Productivity Commission Report – Australian Government Oct 2004.]



Note: In 1991 only 35% of the NSW native forest area was available for future logging – proportions vary in other states [Source: ABARE, 1992]

5b Compile a list of reasons why native forests and woodlands have been cleared.

6 To preserve our native forests and ensure the supply of timber, more use will need to be made of plantation timber. Use information in the table and graph above to write an extended response on the forest timber industry. Include one section on ownership and how you expect it will change, and one on the distribution between the states of native forests and plantations. In your response explain the difference between the categories of ‘native forests’ and ‘plantations’.

8 SOILS AND SALINITY

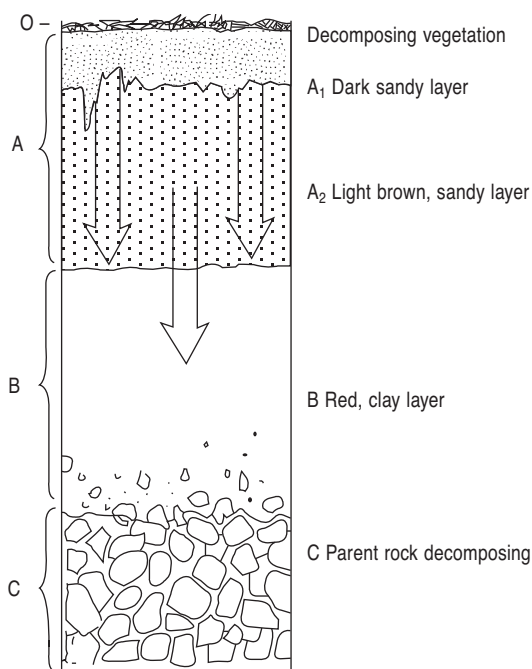
Introduction Soils are one of our most important, yet most vulnerable resources. They are a vital link in the food chains of terrestrial ecosystems and impact greatly on the productivity of our agriculture. Soil degradation costs our farmers \$2.5 billion per year in lost production as they battle infertility, erosion, salinity, acidity, moisture – repelling and soil structure problems. The 1990s were dedicated as the Decade of Land Care to focus on the importance of the problem and stimulate remediation programs.

Terminology

1 Complete the following passage using the terms provided.

topsoil, horizons, alluvial, replenish, organic, intensive, soluble, lithosphere, reactions, leached, weathered, volcanic, salinity, profile

Soil comprises a very thin layer of the (1) _____ that is composed of (2) _____ rock, or parent material and decayed (3) _____ matter called humus. These materials are slowly modified over time in chemical and organic (4) _____ involving water and micro-organisms until (5) _____ or layers develop with increasing depth through the soil (6) _____. As water percolates downward through soil it leaches (7) _____ minerals and moves the smaller particles to deeper horizons, resulting in lower fertility of the (8) _____. Australian soils are mostly (9) _____. When the ice ages provided a new supply of minerals to (10) _____ many Northern Hemisphere soils, Australia was a dry and barren windswept environment. Their age has also led to high levels of (11) _____ and acidity, as salts have accumulated and minerals have been (12) _____ over long periods of time. In limited areas of the east, more recent (13) _____ activity has produced richer basaltic soil, and some river valleys have more fertile (14) _____ soils suitable for (15) _____ agriculture.



2 The diagram above is best known as a

- A cross section
- B soil horizon
- C soil profile
- D soil graph

3 The layers in the soil marked O, A, B and C are

- A levels
- B profiles
- C horizons
- D strata

4 Which of the following is **least** likely to explain why the A1 horizon would be dark and sandy?

- A humus would accumulate there and is usually dark in colour
- B the parent rock is dark and sandy
- C water would move smaller particles downward leaving coarse material behind
- D both a and c

5 The arrows pointing downward through the soil would represent

- A the movement of soil water under gravity
- B the leaching of soluble minerals
- C the movement of fine soil particles
- D all of the above

Multiple choice

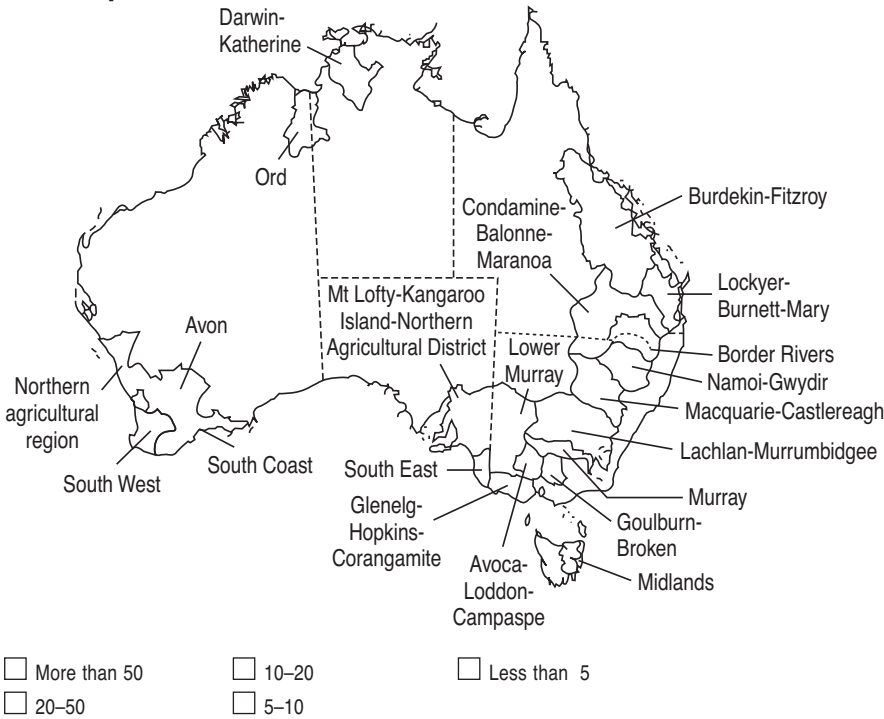
Circle the letter to the answer that best suits the questions following and on the next page.

- 6 In arid climates of Australia
- A soils would be fertile due to the absence of leaching
 - B there would only be sandy particles in the soil profile
 - C salts would accumulate in the upper layers due to evaporation
 - D plant matter would rapidly decompose into the soil

- 7 Salinity in soils can be caused by
- A the presence of salts in parent material
 - B being blown onto the land from sea-spray
 - C being added by saline run-off or dissolved by irrigation
 - D all of the above

Proportion of farms affected by salinity, 2002

[Source: 'National Action Plan for Salinity and Water Quality', ABS, 2002]



Farms with land showing signs of salinity – May 2002

Farm type	Number
Irrigated farms	
Dairy	1005
Beef and/or Sheep	698
Grapevines	504
Grain	411
Mixed, Crop-Livestock	364
Vegetables	349
Fruit	217
Other irrigated	501
Non Irrigated	
Beef and/or Sheep	5396
Grain	4692
Mixed, Crop-Livestock	4578
Other non-irrigated	864
Total Farms	19 579

[Adapted from ABS Year Book Data 2004]

- 8 Use the data provided in the table below and the base map above to produce a choropleth (colour shaded) map on the salinity problem. Darker or stronger colours should be used for higher values. Add a legend to your map.
- 9 Describe in paragraph form, the pattern shown by your completed map. Include a comment on the reasons why different areas of the country might have been included or left out of the survey, the differences in the problem between different states and territories and where you think irrigation or dryland salinity would be more relevant.

- 10a Use the table above to construct a sector graph (pie chart) to show the number of different types of farms affected by the salinity problem. To calculate the size of each sector, divide each figure by the total and multiply by 360°.
- 10b Comment on the pattern shown in your completed graph.

Proportion of farms affected by salinity, 2002

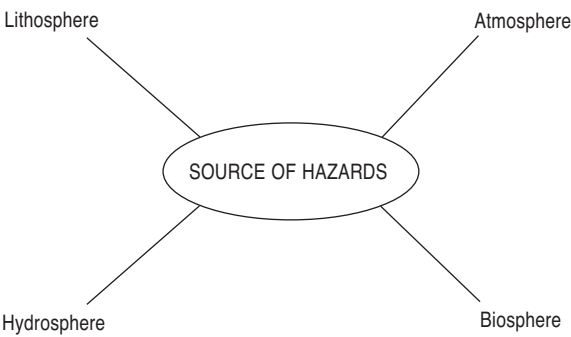
More than 50%	20–50%	10–20%	5–10%	Less than 5%
∞ Northern Agricultural Region ∞ Avon ∞ South Coast	∞ South West ∞ Mt Lofty-Kangaroo Island-Northern Agricultural District ∞ Avoca-Loddon-Campaspe	∞ South East Guelgel-Hopkins-Corangamite ∞ Lower Murray ∞ Midlands ∞ Lachlan-Murrumbidgee	∞ Ord ∞ Namoi-Gwydir ∞ Macquarie-Castlereagh ∞ Murray ∞ Goulburn-Broken	∞ Darwin-Katherine ∞ Burdekin-Fitzroy ∞ Lockyer-Burnett-Mary ∞ Condamine-Balonne Maranoa ∞ Border Rivers

9 NATURAL HAZARDS

Introduction Australia experiences relatively low loss of life from natural hazards, but they have caused approximately 500 deaths and an annual loss (excluding drought) of approximately A\$1.25 billion over the last 25 years. Hazards originate from different parts of our environment, and vary in their nature, impacts and the extent that we can protect ourselves from them.

Hazard sources

- 1 Refer to the mind map at right.
- a Define each of the four 'spheres'.
-
-
-
-
-
-
-
-
-

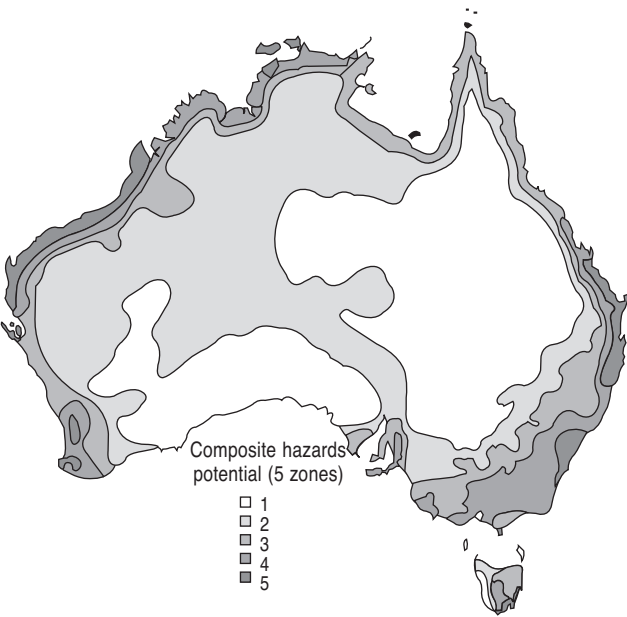


- b Make a copy of the mind map and expand it with examples of different types of natural hazards.

Hazard risk

- 2 Refer to the map at right. Write a paragraph describing the pattern of hazard risk in Australia. Which hazards and factors do you consider would most influence the pattern shown? Explain.
-
-
-
-
-
-
-
-
-
-

Natural hazards potential map based on nine hazards, 1 = Low, 5 = High



[Source: Natural Hazard Resource Centre, Macquarie University]

Who am I?

3 Match the following hazards with the descriptions provided.

Ash Wednesday Bushfires, Cyclone Tracy, Hobart Bushfires, Sydney Hailstorm, Newcastle Earthquake, Thredbo Landslide, Meckering Earthquake, Great Influenza Epidemic, Southern Australian Heatwave

	Description	Event
a	I was a strong hazard of my type but caused little damage due to my remote, sparsely settled location (1968).	
b	I caused extensive damage and death across two southern states in 1983.	
c	Only one person survived me, but I killed 18 others.	
d	I am a biological hazard that killed thousands of people in the winter of 1918.	
e	I caused extensive damage to an industrial city in late 1989, and caused 13 deaths.	
f	I proved that being our “cool capital” was no protection from me in 1967 and 1982	
g	Because of me, building regulations in this city were changed after my visit.	
h	In 1939, I managed to kill 438 people (mostly young and old), without doing much physical damage at all	
i	In January 1994, I arrived at a large east coast city in a severe storm that put quite a dent in insurance company profits.	

Damage to buildings in Australia from natural hazards 1900–99

Tropical cyclones	33%	Hail storms	6%
Floods	22%	Earthquakes	6%
Bushfires	19%	Tornadoes	4%
Wind storms	9%	Landslides	1%

[Source: Natural Hazards Research Centre, Macquarie University]

4 Use the data in the table above to construct a sector graph or divided bar graph in the space below.

5 Answer True or False to the following:

a A natural hazard is a naturally occurring event that has the potential to harm human life, health, property or income.

b Australia has a relatively low risk from earthquakes as it is situated on a stable crustal plate and not over major fault lines.

c Tropical cyclones are a major hazard in southern Australia in summer.

d Bushfires caused the most building damage of any natural hazard in Australia last century.

d A natural hazard is the same as a natural disaster.

e All natural hazards that cause deaths also cause substantial property damage.

6 Write a brief description of a recent disaster caused by a natural hazard in Australia.

10 DROUGHT AND BUSHFIRES

Introduction Australia is particularly prone to the hazards of drought and bushfire. Our location under a zone of eastward migrating highs or anticyclones and the El Nino phenomenon combine to limit our rainfall and make drought a frequent problem. Our highly flammable natural vegetation is particularly vulnerable to fires under these conditions. Drought and bushfires vary in terms of their specific causes, spatial extent, duration, the degree to which we can predict and prepare for them, and the effects they have on our people, economy and environment.

Australia's low rainfall

1a Explain how high pressure systems and the El Nino effect cause low rainfall in Australia. You could use diagrams as your explanation.

1b What other factors contribute to low rainfall in parts of Australia?

The 2002–03 drought

Rainfall for April 2002–January 2003

- Rainfall percentiles
- 1

0–5 Severe deficiency
- 2

5–10 Extremely low (rainfall)
- 3

10–20 Well below average
- 4

20–30 Below average
- 5

30–70 Average
- 6

70–80 Above average
- 7

80–90 Well above average
- 8

90–100 Extremely high (rainfall)



2 Although conditions had eased in some areas by 2005, the continuing 2002 drought was still affecting many parts of Australia and was being described as the worst on record.

a Complete the choropleth map above by colouring the zones. Use darker to lighter warm colours for 1–4, blue for 8, greens for 7/6, and leave 5 blank.

b Write a paragraph describing the extent and severity of the drought for the period shown. Specific reference could be made to particular states or regions.

3 Construct a table like the one commenced at right on the effects of drought and add as many items as you can to it. Some suggestions are provided in the box above the table. Your features could be grouped into sections such as rural areas, urban areas, and the environment.

crops / crop production, farms costs / incomes, farm product prices, country town businesses, transport services, farming families, government reaction, urban water supply, natural vegetation, native animals, soils, river/water quality, other ideas

Feature	Impacts/Changes	Explanation
Farm animals (eg. sheep and cattle)	fall in number	lack of feed, farmers sell off stock, animals die, reduced breeding activity

4 We cannot prevent droughts, and prediction is still unreliable, but we can prepare for and manage the effects of droughts. The federal government's Bureau of Rural Science reports the following as being the most common rural management strategies for major droughts.

Percentage of farmers

handfeeding 1%, adjustment (sending stock away) 1%, storing grain 2%, reducing spending 2%, obtaining off-farm employment 3%, building water storages 9%, improving irrigation efficiency 13%, destocking 18%, storing fodder 28%

- 4a Construct a multiple bar graph to show these figures.
- 4b Indicate with a symbol or colour code if you consider them to be preparation for, or response to the drought. Could some be both?
- 4c Explain briefly how each one would help.

Bushfires

5 The 2002–03 bushfire season was one of the worst Australia has experienced. Add the totals in each column in the table below to get the national figures.

Bushfire statistics – 2002–03 season

	No. of bush-fires attended	Area burnt '000 ha	Fatalities	Houses destroyed	Stock lost
NSW	459	1465	3	86	3400
Victoria	843	1346	0	43	11000
Queensland	2618	1056	0	6	n.a.
South Australia	1419	50	0	4	27
Western Australia	656	2112	0	n.a.	n.a.
Tasmania	n.a.	52	0	6	n.a.
Northern Territory	n.a.	15000	0	n.a.	n.a.
ACT	n.a.	160	4	501	n.a.
Total					

[Source: AusStats Year Book Australia 2004]

Multiple choice

Circle the letter of the answer that best suits each of the questions 6–10.

- 6 Which of the following had the largest number of fires?
- A New South Wales B Victoria
- C Queensland D Western Australia

- 7 Which of the following had the largest area burnt?
- A New South Wales B Victoria
- C Western Australia D Northern Territory
- 8 Which of the following had the largest fires by average area burnt per fire?
- A New South Wales B Victoria
- C Queensland D Western Australia
- 9 In which of the following would you expect fire related insurance claims to be the highest
- A New South Wales B Queensland
- C Tasmania D ACT
- 10 The State/Territory with its agricultural areas most affected would have been
- A New South Wales B Victoria
- C South Australia D ACT

- 11a Calculate the percentages for each cause of fire listed and for the areas burnt in the table below.
- 11b Write an extended response on the causes, impacts and nature of bushfires. Include the effects that drought and weather conditions such as temperatures, winds, humidity levels and precipitation can have on a fire. What importance can be attached to where a fire occurs?

Causes of Victorian bushfires 1976–96

Fire cause	Average no. per year	Proportion of fires %	Average area burnt	Proportion burnt %
Lightning	149		53096	
Deliberate	145		15649	
Agriculture	96		7799	
Campfires	59		1466	
Cigarettes etc.	41		444	
Unknown	37		2974	
Misc. +	26		10009	
Machinery	15		2551	
Prescribed burning escapes	9		5274	
Public utilities	7		16256	
TOTAL	584	100	115518	100

[Source: AusStats Year Book Australia 2004]
+ includes fires from burning buildings, fireworks etc.

11 DEMOGRAPHIC CHARACTERISTICS

Introduction Australia’s demographic characteristics combine to make Australia unique. Together they influence the nature and identity of Australian society – a multicultural country, with a relatively small population concentrated on the coast and ageing by world standards, although young compared with some developed countries.

Definition **median age** – the age at which half the population is older and half the population is younger

Australia’s population size and growth

- 1a** Use the data below to construct a line graph to show the increase in Australia’s population between 1901 and 2002.
- 1b** Describe the trend that the graph reveals.

Year	Population
1901	3,788.1
1912	4,653.7
1922	5,569.9
1932	6,576.8
1942	7,180.7
1952	8,636.5
1962	10,742.3
1972	13,303.7
1982	15,184.2
1992	17,494.7
2002	19,662.8

[Source: ABS Yearbook Australia Population 1301.0 – 2004]

Population distribution

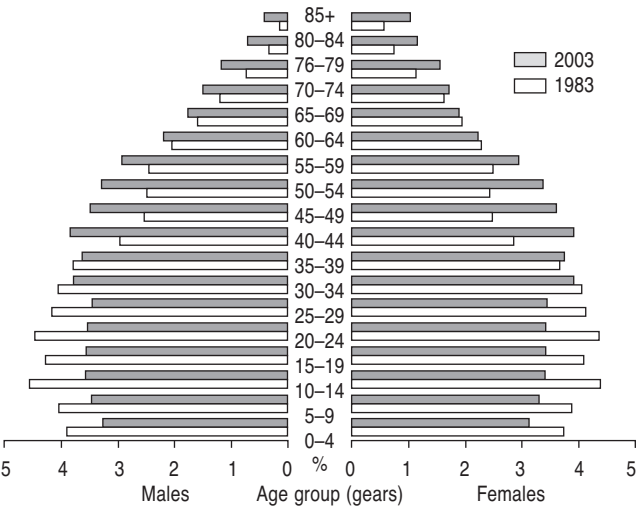
- 2** Complete the following table by filling in the percentage share of Australia’s population of each state and territory.

	Population size '000	Percentage share
NSW	6,640.4	
Vic	4,872.5	
Qld	3,707.2	
SA	1520.2	
WA	1,927.3	
TAS	472.7	
NT	198.0	
ACT	321.8	
AUST	19,662.8	

[Source ABS Yearbook Australia Population 1301.0 – 2004]

Age structure

Australia’s population profile



[Source: Australian Bureau of Statistics–3201.0 Population by Age and Sex, Australian States and Territories]

- 3** Compare the shape of the population profile at left for Australia in 1983 with the shape of the one for 2003. Your comparison should:
- a** Outline the main changes that have taken place.
- b** Suggest possible reasons for the changes.
- 4** Use an atlas, textbook or the Internet to research two recent population pyramids – one from a developed country and another from a developing country. Compare each of these with Australia’s age structure in 2003.
- 5** List some possible consequences of the trends in Australia’s age structure.

	1901 '000	1954 '000	1981 '000	2001 '000
United Kingdom and Ireland	679.6	664.2	1 120.9	1 182.8
New Zealand	25.8	43.4	160.7	394.1
Italy	5.7	119.9	275.0	238.5
Vietnam	n.a.	n.a.	40.7	169.5
China (excluding SARS and Taiwan Prov.)	29.9	10.3	25.2	157.0
Greece	0.9	25.9	145.8	132.5
Germany	38.4	65.4	109.3	117.5
Philippines	0.7	0.2	14.8	112.2
India	7.6	12.0	41.0	103.6
Netherlands	0.6	52.0	95.1	91.2
Malaysia	n.a.	2.3	30.5	87.2
South Africa	0.5	6.0	26.5	86.9
Lebanon	n.a.	3.9	49.4	80.0
Hong Kong	0.2	1.6	15.3	75.2
Poland	n.a.	56.6	59.0	67.5
Yugoslavia	n.a.	22.9	148.6	64.0
United States of America	7.4	8.3	28.9	59.0
Sri Lanka	0.6	2.0	16.8	58.6
Total overseas born	865.5	1286.5	3 128.1	4 482.1
Australia	2 908.3	7 700.1	11 388.8	14 931.2
TOTAL POPULATION	3 773.8	8 986.5	14 516.9	19 413.2

6 Statistics on the countries of birth of Australia's population provide us with part of the picture about the ethnic composition of Australia's population. Suggest other ways by which we could gain a picture of the ethnic composition of the Australian population.

True or false?

7 Label the following statements true or false. Rewrite the incorrect ones correctly.

a Australia's population is ageing.

f Since 1981, there has been an increase in the overseas born population from all the Asian countries listed in the table.

Multiple choice

A Vietnam **B** China
C Philippines **D** India

A unique population

13 Write a description of Australia's demographic characteristics for someone who has never been to Australia. Your description should include information about Australia's: population size and growth, population distribution, changing age structure and changing ethnic composition.

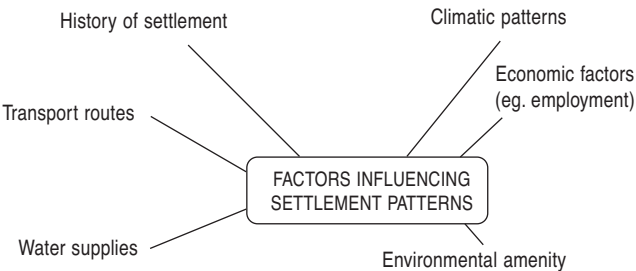
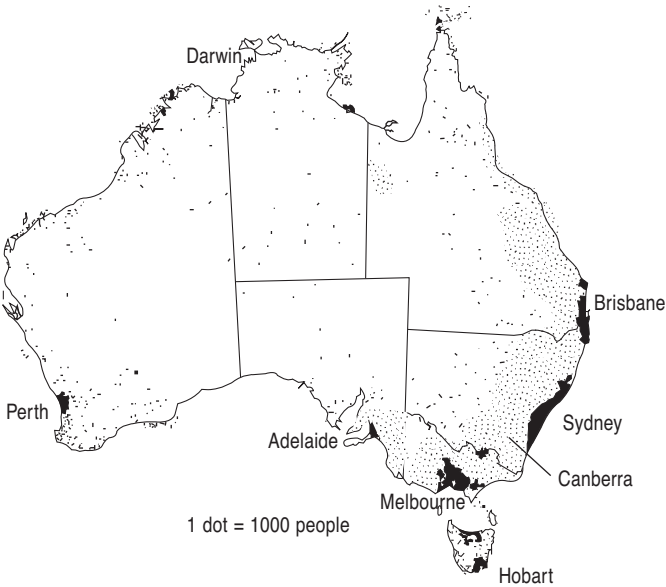
12 POPULATION DISTRIBUTION

Note to teacher

Students will need to use an atlas and a blank outline map of Australia to complete this worksheet.

Introduction Australia is less evenly populated than many other countries due to a variety of factors. There is an average of 2.6 people per square kilometre. Much of the country is very sparsely settled. At one extreme, 250 Statistical Local Areas have less than one person per square kilometre. Density is highest around our capital city centres, and reaches 8000 persons per square kilometre in inner Sydney. Very high and very low densities can present problems for residents of such areas and high densities also usually create significant environmental impacts.

- 1 Refer to the map below and write a paragraph describing the pattern of Australia's population distribution.
- 2 Make a copy of the mind map below. Expand it by adding points, then write a paragraph using each section to help explain our population distribution pattern.



Distribution by distance – percentage of population living in specified zones

State or territory	Major cities	Inner regional	Outer regional	Remote	Very remote	Total population '000	Notes
NSW	71.5	20.5	7.3	0.6	0.1	6575.2	(i) Tasmanian and NT cities do not reach the size required for major city status and are therefore classified as inner or outer regional. (ii) Regional and remote areas are based on distance from the nearest major city (iii) Total population excludes other territories
Vic	73.2	21.2	5.5	0.1		4804.7	
Qld	52.3	25.9	17.8	2.5	1.5	3628.9	
SA	71.8	12.4	11.8	3.0	1.0	1511.7	
Western Australia	70.6	12.2	9.8	4.8	2.6	1901.2	
Tasmania		63.5	34.1	1.8	0.6	471.8	
NT			54.2	21.2	24.6	197.8	
ACT	99.8	0.2				319.3	
AUSTRALIA	66.3	20.7	10.4	1.7	0.9	19 410.6	

[Adapted from ABS 2004 Year Book – Census Data 2001]

3 Use the data in the table on the previous page. Calculate a sector size (as for a pie graph) to represent the total of population of each state and territory. Draw a sector for each state around the country on your map of Australia as close as possible to the relevant state/territory, using the same length radius.

Multiple choice

Use the map on the previous page and the table at right to answer the following questions. Circle the letter to the answer that best suits each of the questions 3–9.

- 4 Australia is least densely populated in the
- A Northeast B Southeast
C Southwest D Northwest
- 5 Which of the following states/territories has the lowest proportion of people living in major cities.
- A New South Wales B Queensland
C Victoria. D South Australia
- 6 Which of the following states has the highest proportion of people living in both inner regional and outer regional areas
- A Western Australia B South Australia
C Victoria D Queensland
- 7 The state or territory with the most geographically concentrated population is
- A South Australia B ACT
C Western Australia D Northern Territory
- 8 The state or territory with the largest proportion of people living in remote areas is
- A South Australia B Queensland
C Western Australia D Northern Territory
- 9 Which state or territory would have a major city population of 1,897,914 people?
- A Queensland B Western Australia
C South Australia D Victoria.
- 10 Identify the reasons why people would live in remote communities and outline some of the difficulties they would experience.
- 11 Use the 2002 population figures in the table above to work out a classification scheme for the cities shown. Locate them on your map of Australia, using a different symbol or colour for each size group. Add a legend to your map.

	30 June 1997 '000	30 June 2002 '000	Change %
Capital City			
Sydney	3928.7	4170.9	1.2
Melbourne	3309.6	3524.1	1.3
Brisbane	1543.8	1689.1	1.8
Perth	1316.3	1413.7	0.6
Adelaide	1083.9	1114.3	1.4
Greater Hobart	196.0	198.0	0.2
Darwin	98.9	107.4	1.7
Canberra	308.7	321.4	0.8
District			
Newcastle	469.3	497.5	1.2
Gold Coast– Tweed	368.5	439.7	3.6
Canberra– Queanbeyan	346.7	365.4	1.1
Wollongong	258.4	272.1	1.0
Sunshine Coast	162.4	191.9	3.4
Geelong	152.8	161.7	1.1
Townsville	123.6	137.4	2.1
Cairns	108.5	114.5	1.1
Toowoomba	103.1	111.4	1.6
Albury–Wodonga	93.6	99.3	1.2
Launceston	98.7	99.1	0.1
Ballarat	79.7	84.8	1.3
Bendigo	75.2	80.9	1.5
Burnie– Devonport	78.8	77.4	–0.4
Bathurst–Orange	72.2	76.5	1.2
La Trobe Valley	75.8	74.9	–0.2
Mackay	61.7	65.8	1.3
Rockhampton	64.3	64.2	–
Mandurah	52.1	62.0	3.5
Bundaberg	54.7	57.7	1.1
Wagga Wagga	51.7	52.5	0.3
Bunbury	43.3	50.8	3.3
Coffs Harbour	43.1	47.1	1.8
Mildura	41.9	46.2	2.0
Shepparton	41.6	45.7	1.9
Tamworth	40.7	42.7	1.0

- 12 Use the data in the table above:
- a Identify the 5 fastest growing centres in order and locate them on your map.
- b What do they have in common?
- c Why do you think they might be growing so fast?
- d What problems could this rapid growth create?
- 13 Write an extended response on the advantages and problems of living in large Australian cities. What can be done to solve or reduce the problems?

13 INDIGENOUS AUSTRALIANS

Introduction Despite the fact that Australia is a wealthy country, many of its Indigenous people are living in very poor conditions. Lack of access to health, education and other basic services such as roads and water along with high levels of unemployment has led to low life expectancy and population pyramid that is more typical of a developing country.

Definitions

- fertility rate** – the average number of children per woman
- life expectancy at birth** – the average number of years a person born at a particular time can expect to live

1 In your own words define the term ‘Indigenous’. Apply this term in a sentence relating to Australia’s population.

Population growth

Australia’s indigenous population

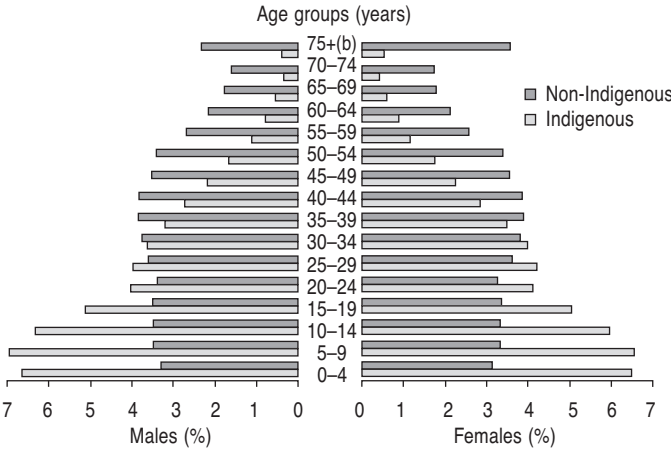
Year	Population
1966	80 207
1971	115 953
1976	160 915
1981	159 897
1986	227 593
1991	265 371
1996	352 970
2001	410 003

[Source: Census of Population and Housing; Commonwealth Bureau of Census and Statistics in ABS 2004 Yearbook]

- 2a On a separate sheet construct a line graph to show the increase in the Indigenous population from 1966 to 2001.
- 2b During which 5-year period did Australia’s Indigenous population experience the greatest population increase?
- _____
- 2c Suggest reasons for the sharp increase in the size of the Indigenous population during this period.
- _____
- _____
- _____
- _____
- _____

Age structure

Age distribution of Indigenous and non-Indigenous population at 30 June 2001



[Source: ABS 2003]

(a) Estimated resident population.

(b) The 75+ age group includes all ages 75 and over and therefore is not strictly comparable with five-year age groups in the rest of this graph.

- 3a Calculate the proportion of the following aged over 65:
- Indigenous Australians
- non-Indigenous Australians
- 3b Account for the differences in the proportion of Indigenous and non-Indigenous Australians aged over 65.
- 3c Calculate the proportion of the following aged under 15:
- Indigenous Australians
- non-Indigenous Australians
- 3d Account for the differences in the proportion of Indigenous and non-Indigenous Australians aged under 15.

Life expectancy

4b Suggest reasons for the huge disparities in life expectancy between each of these groups.

	Indigenous Australians		Non-Indigenous Australians	
2001	Males	Females	Males	Females
Life expectancy at birth	56	63	77	82
Fertility rate		2.1		1.7

[Source: ABS Yearbook Australia 2004]

4a Calculate the difference in life expectancy for Indigenous and non-Indigenous males and females.



True or false?

5 Label the following statements true or false. Rewrite the incorrect statements correctly.

	Statement	T or F?	Correct statement
a	Australia's Indigenous population increased by 16 percent between 1996 and 2001.		
b	Indigenous Australians have an older population structure than non-Indigenous Australians.		
c	The largest age group for Indigenous Australians in 2001 was the 0–4 age group.		
d	The life expectancy of non-Indigenous Australian females is 17 years greater than the life expectancy of Indigenous Australian females.		
e	A major factor explaining the disadvantage of many Indigenous Australians is the fact that a large proportion of them live in isolated areas that are not well serviced.		

6 Use your notes, text and the Internet to make comparisons between Indigenous and non-Indigenous Australians in the following areas.

	Indigenous Australians	Non-Indigenous Australians
Housing		
Health		
Education		
Employment		

14

NATIVE TITLE

Introduction Native Title is the recognition under Australian law that indigenous people prior to European settlement had a system of law and ownership of their land. This recognition finally came about in 1992 as a result of the High Court's Mabo decision. Until this time the doctrine of terra nullius had prevailed.

Terminology

1 Match the following terms with the correct definition.

coexistence, Crown land, freehold land, Mabo, mediation, Native Title, Native Title Tribunal, pastoral lease, terra nullius, Wik		
a	When all people with an interest in the area covered by a Native Title claimant application are brought together by the Native Title Tribunal to discuss their own interests in that area and the interests of others.	
b	The decision of the High Court of Australia in 1992 which recognised that the Meriam people had Native Title over their lands.	
c	The existence and exercise of Native Title rights alongside the rights of others to areas of land or waters.	
d	Land belonging to the government.	
e	The rights and interests of Aboriginal and Torres Strait Islander people in land and waters according to their traditional laws and customs that are recognised under Australian law.	
f	Outright land ownership.	
g	An independent body established under the <i>Native Title Act 1993</i> to assist people to resolve Native Title issues.	
h	Land leased from a state government or the Northern Territory government for the purposes of pastoral activity.	
i	The idea that the Australian continent belonged to no one at the time of European arrival.	
j	The decision of the High Court of Australia in 1996, which states that the grant of a pastoral lease does not necessarily extinguish Native Title and that Native Title rights may coexist with the rights of some leaseholders.	

Recognition of Native Title

2 In the table below place each of the following events next to the date which they occurred and in the third column outline the impact each one has had on Aboriginal people.

Creation of the Native Title Act, Establishment of the Native Title Tribunal, High Court's Mabo Decision, High Court's Wik Judgement		
	Event	Impact
3 June 1992		
1993		
1994		
December 1996		

Claiming Native Title

3 Tick whether Native Title could possibly be claimed in the following locations or not. In the last column, briefly state why Native Title could be possible or no.

	Situation	Possibly	Definitely not	Explanation
a	a primary school in suburban Brisbane which has been operating for 87 years			
b	pastoral land where Aboriginal people have had a continuing connection with the land and waters in accordance with traditional laws and customs			
c	an individually owned freehold title house in Adelaide			
d	vacant crown land on the outskirts of Townsville			
e	a shopping centre in Perth covered by a commercial lease			

Native Title determinations

A Native Title determination is a decision that Native Title does or does not exist in a particular area of land or waters. The Federal Court of Australia, the High Court of Australia or another recognised body makes determinations. At the start of 2005 there had been 54 Native Title determinations in Australia. In 35 of these it was determined that Native Title existed in the entire determination area or in parts of the area and in 19 it was determined that Native Title did not exist in the determination area.

4 Label the following statements true or false. Rewrite the incorrect statements correctly.

	Statement	T or F?	Correct statement
a	Native Title was first recognised in Australia in 1989 when the High Court recognised that the Meriam people of Arnhem land had Native Title over their traditional lands.		
b	The Wik decision found that pastoral leases do not necessarily extinguish Native Title.		
c	Determinations about Native Title are made by the Native Title Tribunal.		
d	To date 65 percent of Native Title claims have been successful or partially successful.		
e	Native title cannot be claimed on freehold land.		

Summary

5 Complete the summary by filling in the correct term in the spaces provided.

Federal Court, Mabo, mediate, Native Title, Native Title, terra nullius, 35, 54, 1992

When Europeans came to Australia in 1788 they declared the land (1) _____. It was not until (2)_____ that this doctrine was overturned as a result of the (3) _____ decision. In 1993 the Commonwealth Parliament passed the (4) _____ Act, which established the (5) _____. _____ Tribunal. The role of the Tribunal is to (6) _____ Native Title claims under the direction of the (7) _____ of Australia. By 2005 there had been (8) _____ Native Title determinations and in (9) _____ of these Native Title has been determined to exist or partially exist.

15

AUSTRALIA'S CHANGING AGE STRUCTURE

Introduction As **life expectancy** increases and **fertility rates** decline, Australia is experiencing an ageing of its population. This ageing of the population presents a huge challenge for planners at all levels of government in the 21st century.

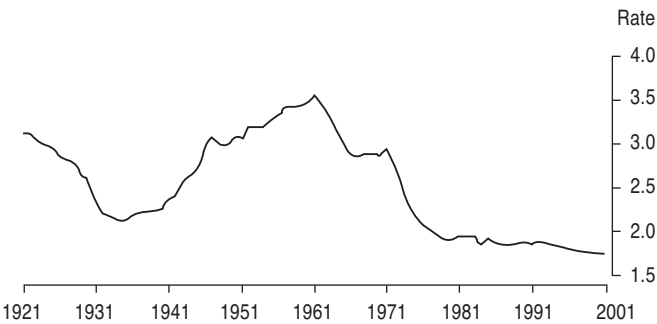
Definitions

fertility rate – the average number of children per woman

life expectancy at birth – the average number of years a person born at a particular time can expect to live

Declining fertility rate

Total fertility rate, Australia



[Source: ABS Yearbook 2004]

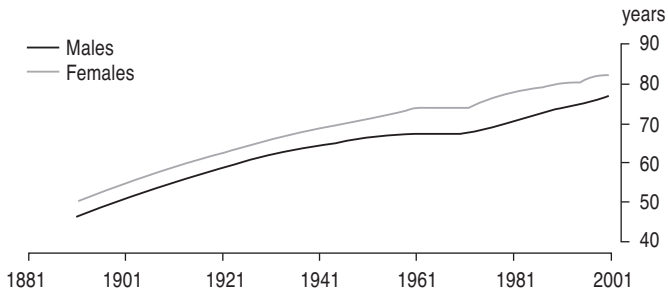
1 Complete the table at right by recording the fertility rate for each of the years shown and suggesting possible reasons for the fertility rate in each of these years.

Year	Fertility rate	Possible reasons
1921		
1934		
1948		
1966		
1979		
2001		

Increasing life expectancy

- 2 Refer to the graph at right.
- a Calculate the increase in life expectancy for males and females between 1901 and 2001.
- Males _____
- Females _____
- b Suggest reasons for changing life expectancy during this period.

Life expectancy at birth, Australia



[Source: ABS Yearbook 2004]

Multiple choice

[Source: ABS Yearbook Australia 2004]

- ∞ Calculating the increase in the Australian population 1901–2002.
- ∞ Calculating the percentage change in the proportion of the Australian population in age group shown from 1901 to 2002.
- ∞ Calculating the increase in the median age of Australia's population between 1901 and 2002.



4 Australia's fertility rate experienced the greatest increase between:

- 5** Australia's fertility rate experienced the greatest decline between:

- A** 1921–31 **B** 1941–51
C 1961–71 **D** 1991–2001

6 Australia experienced the greatest increase in life expectancy between:

- A** 1901–21 **B** 1921–41
C 1941–61 **D** 1981–2001

7 Increase in Australia's life expectancy throughout the 20th century, can best be explained by:

- A** improved medical services
B better nutrition
C better water supply and sewerage systems
D all of the above

8 The age group to experience the largest change in its share of the population between 1901 and 2002 was:

- A** 0–14 **B** 15–64
C 65–84 **D** 85 and over

9 Consider the responsibilities of each level of government and make a list of the possible ways each level of government can continue to and further improve their catering for an ageing population.

16

Students will need to use an atlas and a blank outline map of Australia to complete this worksheet.

Multiple choice

- 2** Examine the table on the previous page then circle the answer that best suits the question.
- a** The single most valuable agricultural product in 2001–02 was
- | | |
|----------------|-----------------|
| A wheat | B sheep |
| C wool | D cattle |
- b** An agricultural product that has declined in value in the period shown is
- | | |
|---------------|--------------------|
| A oats | B sugarcane |
| C pigs | D wool |
- c** The product that has risen most in value is
- | | |
|-----------------|-----------------|
| A cattle | B sheep |
| C milk | D grapes |
- d** The product that has risen least in value is
- | | |
|---------------|--------------------|
| A oats | B sugarcane |
| C wool | D eggs |
- e** The four most valuable agricultural products in descending order are
- | |
|-------------------------------------|
| A wool, wheat, cattle sheep |
| B wheat, cattle, milk, wool |
| C cattle, wheat, milk, wool |
| D wheat, sheep, wool, cattle |
- f** The category to increase most in value was
- | | |
|-----------------------------|----------------------------|
| A crops | B livestock |
| C livestock products | D total agriculture |

Manufacturing centres

Manufacturing involves the transformation of raw materials into finished products and fuels. Most capital cities dominate the manufacturing in their state, although there are some specialised industrial cities and some towns that process minerals and agricultural goods. Australia's manufacturing is divided between the states:

- NSW 32% • Victoria 32% • Queensland 15%
- Western Australia 9% • South Australia 7%
- Tasmania 3% • ACT 1% • Northern Territory 1%.

3a Show the state shares of manufacturing on your map of Australia. You could use pie sectors, a line/bar graph or small squares on graph paper each representing 1%.

3b The major manufacturing centres are Gladstone, Brisbane, Newcastle, Sydney, Wollongong, Melbourne, Geelong and Kwinana. Secondary centres include Cairns, Townsville, Mackay, Bundaberg, Toowoomba, Gold Coast, Tamworth, Dubbo, Orange, Bathurst, Canberra, Wagga Wagga, Albury-Wodonga, Shepparton, Ballarat, Bendigo, Warrnambool, Portland, Mt. Gambier, Adelaide Fremantle Perth and Darwin, Hobart, Burnie and Devonport. Locate these places on your map using two different symbols or colours.

Where am I?

- 4** Match the industrial locations with the clues **a–o** in the right column.

Dubbo, Mackay, Kwinana, Newcastle, Shepparton, Brisbane, Geelong, Canberra, Wollongong-Port Kembla, Toowoomba, Adelaide, Burnie, Mt Isa, Gladstone, Melbourne

- a** I am located a short distance south of a state capital and have oil refining and steel industries.
- b** I am an inland town well known for mining and processing of copper and other valuable metals.
- c** Mitsubishi motor cars are (or were) made here.
- d** My main manufacturing industry is the milling of sugar cane.
- e** Most of my industry is light so it does not pollute my special environment.
- f** The closure of steelworks has led to a restructuring of industry here.
- g** Nearby coal deposits provide cheap electricity needed aluminium production.
- h** The farmlands known as “The Darling Downs” provide livestock for my major industry of food processing.
- i** I am a state capital on a river which has oil refineries located on each side at the river mouth.
- j** I am a city with a very large amount of diverse industry and the port which handles the largest value of goods in Australia.
- k** Although I now have an oil refinery, car plant and other major industries, my first major industry was wool processing.
- l** I am a rapidly growing country town about 300 km NW of my state capital and have a pattern of general light industry and food processing.
- m** My main industry is processing the timber brought from my hinterland.
- n** It was the local coal deposits that attracted the iron and steel industry to locate here.
- o** I am located in the Goulburn Valley where fruit growing keeps my canning and fruit processing industries well supplied.

17

REGIONAL AND GLOBAL LINKS – DEFENCE

Note to teacher

Students will need to use an atlas and a blank outline map of Australia to complete this worksheet.

Definition GDP (gross domestic product) – all the goods and services produced in a country in a year

Location of Australia’s defence force operations

1a Use an atlas to mark Australia’s defence force operations on a map of the world.

- A** Middle East – Operation Catalyst – Australia’s contribution to the rehabilitation of Iraq

B Middle East – Operation Slipper – Iraq and Afghanistan

C East Timor – Operation Citadel and Operation Spire

D Solomon Islands – Operation Node

E Middle East – Operation Paladin – Israel, Syria and Lebanon
- F** Sinai – Operation Mazurka

G Eritrea and Ethiopia – Operation Pomelo

H Southern Ocean – Operation Mistral

I Northern Australia and Sea Air Approaches – Operation Cranberry

J Australian border protection – Operation Relex II

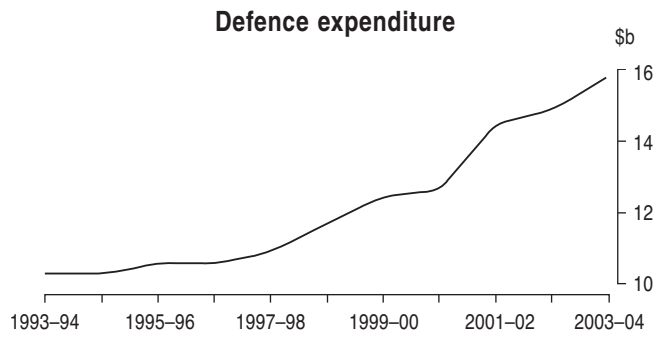
1b Describe and explain the spatial distribution of Australia’s defence force operations.

1c Match the letter for each of the defence force operations listed above with the description of the operation below.

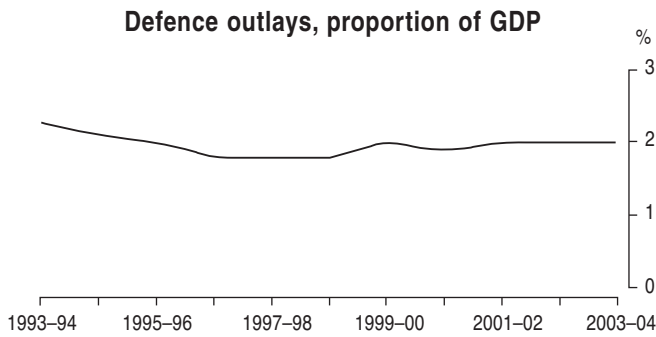
Australia’s defence force Operations, 2004

1 This is Australia’s contribution to the UN Truce Supervision Organisation (UNSTO) It was established in 1948 to supervise the truce agreed at the conclusion of the first Arab/Israeli War.		2 This operation works to detect, intercept and deter vessels carrying unauthorised arrivals from entering Australia through the North-West maritime approaches. It includes navy, airforce and army supported by Coastwatch and Customs.	
3 Australia contributes 2 army officers to the UN Mission in Ethiopia and Eritrea (UNMEE). The UNMEE mission includes monitoring the cessation of hostilities, troop deployments and the temporary security zone between the two countries.		4 Australia has led a Regional Assistance Mission in this country. The Australian contribution has included 450 Australian Defence Force personnel, Australian Federal Police and personnel from the Australian Protective Service.	
5 This is Australia’s contribution to the Multinational Force and Observers (MFO) in the Sinai. 25 Australians assist in the peace process by monitoring the border and preparing daily operation briefings.		6 The purpose of this operation is to detect illegal activity in northern Australia such as smuggling and illegal fishing. The operation is primarily undertaken by RAN Freemantle Class Patrol Boats and Army Reserve Personnel from the Regional Force Surveillance Unit.	
7 During this operation Australia contributed 25% of the UN peacekeeping force. The Australian Defence Forces commitment is continuing here with head-quarters, logistics and engineering and military liaison tasks.		8 This operation is Australia’s contribution to the war against terrorism.	
9 This operation provides support to the civil agencies enforcing Australian sovereign rights and fisheries laws in the Southern Oceans.		10 Australia’s contribution to the rehabilitation of Iraq. The focus is on stabilisation and recovery programs and facilitating the transition to self government. It comprises 880 personnel.	

Defence spending and staffing



[Source: Department of Defence]

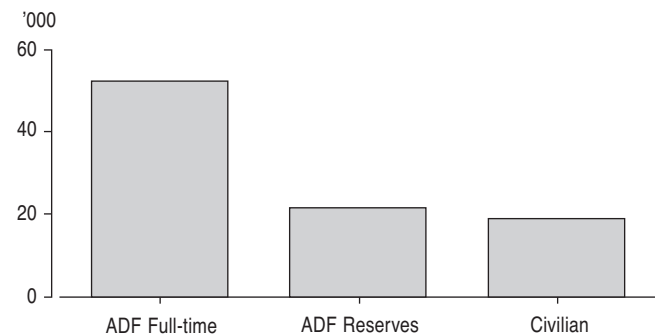


[Source: Department of Defence]

Australian defence force staffing, 30 June 2003

	ADF full time	ADF reserves
Navy	12 864	1 616
Army	25 289	17 172
Air Force	13 638	2 800
TOTAL	51 791	21 588

[Source: Department of Defence]



[Source: Department of Defence]

True or false?

- 2 Read the data above, then label the following statements True (T) or False (F).
- a Defence spending has experienced a decline between 1993-94 and 2003-04.
- b The trend has been for defence spending to increase with GDP.
- c Over 92 000 people are employed in the Australian defence forces.
- d Army personnel represent about 20% of full-time defence force employees.
- e The smallest area of the defence forces in terms of employment is the air force.

ANZUS – a defence treaty

- 3 Insert the correct words from the list below to complete the paragraph on the ANZUS Treaty.
- Article IV, Australia, cornerstone, independent, interests, invoked, mutual aid, New Zealand, Pacific, San Francisco, USA, terrorism, World Trade Centre, 1951
- ANZUS is a defence security pact for the (1) _____ region, between (2) _____, (3) _____ and the (4) _____. The agreement was signed in (5) _____ in September (6) _____. The purpose of the treaty was to provide (7) _____ in the time of aggression and to settle disputes by peaceful means.
- There have been many criticisms of the ANZUS Treaty. Many critics believe that Australia should be more (8) _____. They have argued that Australia cannot rely on the USA to protect Australia's (9) _____.
- The current Australian government sees the ANZUS Treaty as the (10) _____ of Australian Defence policy.

- In September 2001, after the attacks on the (11) _____ the ANZUS Treaty was (12) _____ for the first time. The Australian government in consultation with the USA decided that (13) _____ of the ANZUS Treaty applied to the terrorist attacks on the USA. The Australian government has expressed a commitment to working with the USA in combating (14) _____.

Advantages and disadvantages

- 4 Using your own knowledge, texts and notes list in tabular form the advantages and disadvantages of defence operations and spending to Australia. You should consider the economic, political and social aspects of Australia's defence commitment.

18

REGIONAL AND GLOBAL LINKS – INTERNATIONAL TOURISM

Introduction Tourism is defined by the World Tourism Organisation (WTO), as the activities of people who travel outside their usual environment for no more than a year for leisure, business or other purposes. International tourism to Australia is a vital part of the economy. Its impact can be felt in a whole range of industries including the accommodation, restaurant, transport and retail industries. In 2001–02 tourism directly generated 549 000 Australian jobs.

Incoming tourists

International tourist numbers

Year	'000
1997	3 974
1998	3 859
1999	4 143
2000	4 567
2001	4 475
2002	4 463

- 1a
- Construct a column graph to show the number of international tourists to Australia between 1997 and 2002.
- 1b
- Describe the trend in international tourist numbers to Australia between 1997 and 2002.
- 1c
- Suggest reasons for the drop in international tourist numbers in 2001 and 2002.

International visitors by country of residence

	Visitors '000							
Country of residence	1997	1998	1999	2000	2001	2002	2002 visitor nights	2002 expenditure \$ million
New Zealand	621	641	661	741	736	716	10 366	956
Japan	766	704	662	670	626	668	7 445	909
Hong Kong	137	130	128	141	140	136	3 883	435
Singapore	201	216	234	243	255	246	4 632	682
Malaysia	126	102	126	136	133	140	3 694	385
Indonesia	138	83	82	89	87	80	3 069	362
Taiwan	139	135	134	120	101	89	1 984	209
Thailand	62	45	56	66	72	75	2 464	253
Korea	220	62	100	145	160	172	5 313	548
China	64	73	87	115	148	180	6 812	648
Other Asia	89	92	102	115	114	107	4 767	342
USA	310	353	392	457	418	407	11 471	1 359
Canada	61	68	75	85	89	86	3 238	249
United Kingdom	388	448	509	556	583	605	22 464	1 730
Germany	124	123	140	139	142	131	5 188	467
Other Europe	319	344	389	457	398	385	15 651	1 309
Other countries	209	239	265	291	274	239	6 960	532
TOTAL	3 974	3 859	4 143	4 567	4 475	4 463	119 398	11 377

[Source: Bureau of Tourism Research]

True or false?

- 2
- Use the table above. Label the questions a–e on this page and the next T or F. Rewrite the false statements correctly.
- a
- Between 1997 and 2002 total international visitor numbers to Australia experienced a 10% increase.

- b** In 1997 the largest number of international visitors, were from Japan but by 2002 the largest number of international visitors to Australia were from New Zealand.
- c** New Zealanders comprised 16 percent of the international tourists to Australia in 2002.
- d** People from the UK spent the largest number of visitor nights in Australia in 2002, comprising 8% of the total.
- e** In 2002, the largest visitor expenditure was by people from the UK.

Trends in the data

3 The following trends can be identified from the data on international visitors to Australia. Suggest possible reasons for each trend. Consider global conditions and economic conditions in the relevant countries.

Trend
a An increase in tourist numbers from New Zealand between 1997 and 2002.
b An overall decline in tourist numbers from Japan between 1997 and 2002.
c A dramatic drop in tourist numbers from Korea between 1997 and 1998 and a continual increase since.
d A continual increase in tourist numbers from China during the period 1997–2002.
e A continual increase in tourist numbers from the United Kingdom between 1997 and 2002.

Purpose of visit

4 Use the following statistics to construct a pie graph showing the purpose of visit of international tourists to Australia.

Purpose	Visitors '000
Holiday	2 446
Visiting friends and relative	853
Business	659
Education	262
Employment	59
Other	184
TOTAL	4 463

Multiple choice

- 6** Circle the answer that best suits the question.
- a** The largest increase in international tourist numbers to Australia was experienced between:
- A** 1997 and 1998 **B** 1998 and 1999
C 1999 and 2000 **D** 2000 and 2001
- b** The country which experienced the largest increase in visitor numbers to Australia between 1997 and 2002 was:
- A** New Zealand **B** China
C USA **D** UK

Tourist attractions

5 Draw up a table like the one below. Match the following tourist attractions with the region to which they belong and classify each attraction as physical or human, depending on its qualities.

Airlie Beach, ANZAC Hill, beaches of Coolangatta, Daintree National Park, Fitzroy Gardens, Fraser Island, Fremantle, Glenelg, Great Barrier Reef, Hahndorf, Hamilton Island, Harbour Bridge, Kata Tjuta, John Flynn's grave, Kings Park, Litchfield National Park, Mindil Beach Markets, Mooloolaba, Mount Coottha Lookout, Mount Warning, Opera House, The National Gallery, natural harbour and beaches, Noosa National Park, Queen Victoria Market, South Bank, Surfers Paradise, Taronga Zoo, Uluru, The War Memorial, Warner Brothers Movie World, Werribee Open Plains Zoo, whales

Top 15 regions visited by International visitors in Australia, 2002

Region	Physical	Human
Sydney (NSW)		
Melbourne (Vic)		
Gold Coast (Qld)		
Tropical North (Qld)		
Brisbane (Qld)		
Perth (WA)		
Adelaide (SA)		
Petermann (NT)		
Whitsundays (Qld)		
Sunshine Coast (Qld)		
Hervey Bay/Maryborough (Qld)		
Northern Rivers Tropical (NSW)		
Alice Springs (NT)		
Canberra (ACT)		
Darwin (NT)		

- c** The country which experienced the largest decrease in visitor numbers to Australia between 1997 and 2002 was:
- A** Japan **B** Indonesia
C Taiwan **D** Korea
- d** In 2002, the average number of visitor nights per tourist was largest for tourists from:
- A** Indonesia **B** China
C UK **D** Germany
- e** In 2002 average tourism expenditure per person was highest for people from:
- A** Singapore **B** Korea
C USA **D** UK

19 REGIONAL AND GLOBAL LINKS – MIGRATION

Introduction Australia is a land of migrants, with a diverse multicultural population. For centuries migrants have been coming to Australia from all over the world. Together they have made a huge economic and social contribution to Australia.

Terminology

1 Match the following terms with the correct definition.

asylum seeker, emigration, family migrants, humanitarian migrants, immigration, net overseas migration, pull factors, push factors, skilled migrants

1	The excess of permanent and long term arrivals over permanent and long term departures.	
2	People who have particular work skills, have satisfied a points test, have been nominated by particular employers or have successful business skills and/or significant capital to bring to Australia.	
3	The factors which attract people to another country.	
4	A person seeking refuge.	
5	The movement of people out of a country.	
6	People who have been selected on the basis of a family relationship to a sponsor in Australia essentially spouses, fiancés, dependent children and parents who meet the 'balance of family test'.	
7	People who meet the United Nations Convention definition of a refugee and/or have suffered abuse of human rights, displacement or hardship.	
8	The movement of people into a country.	
9	The factors which force people to leave a country.	

Where do migrants to Australia come from?

Country of birth of settler arrivals

	1982-83		2002-03	
	No.	%	No.	%
China	1 167		6 664	
India	1 673		5 783	
New Zealand	6 867		12 368	
South Africa	2 758		4 603	
United Kingdom	26 444		12 508	
Vietnam	8 690		2 568	
ALL SETTLER ARRIVALS	93 011		93 914	

[Source: ABS Yearbook Australia 2004, Population – International Migration]

2 Complete the table above by filling in the percentage of all settler arrivals which each group comprised in 1982–83 and 2002–03.

Why do people migrate?

3 In the table below, like the one below list the push factors and pull factors for migration

[illegible]

Australia’s migration program

Settler arrivals, by eligibility category

	1996–97	2001–02
Family	36 490	23 344
Skilled	19 697	36 036
Humanitarian	9 886	6 732
New Zealand	17 501	21 458
Other	2 178	1 330
TOTAL	85 752	88 900

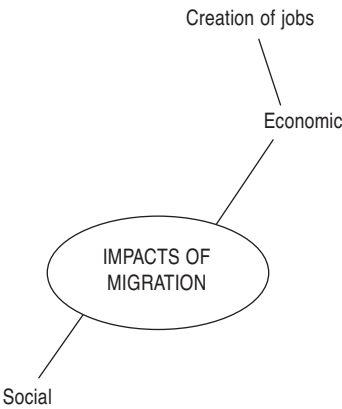
[Source: Department of Immigration and Multicultural Affairs, ‘Immigration Update’ (1996–07 and 2001–02) in ABS Yearbook 2004]

4 Use the table above to help you complete the following paragraph about Australia’s migration program by filling in the blanks below.

Australia’s migration program has three main eligibility categories. (1) _____. (2) _____ and (3) _____. In 2001–02 there were (4) _____ migrants to Australia. This was an increase of (5) _____ people from 1996–7. In 1996–97 the largest eligibility category of migrants to Australia was (6) _____ migrants who made up (7) _____% of the total. However by 2001–02 this group of migrants had declined to (8) _____, comprising (9) _____ of total settler arrivals. During 2001–02 the largest eligibility category of migrants in Australia was (10) _____ migrants. This group had experienced a percentage increase of (11) _____ from 1996–07. Humanitarian migrants experienced a decline of (12) _____ between 1996–97 and 2001–02.

Impacts of migration

5 Create a mind map around the central theme – Impacts of migration. Use the figure below as a starting point.



True and false?

- 6 Label the following statements True or False. Rewrite the incorrect statements correctly.
- a Australia’s migration program has 5 eligibility categories.
 - b During the period 1996–97 to 2002–03 Australia experienced a shift in its migration focus from skilled migrants to family migrants.
 - c In 2002–03 New Zealanders comprised the largest proportion of migrants to Australia.
 - d Better living conditions are a pull factor for migrants.

Multiple Choice

- 7 Circle the letter of the answer that best suits the question.
- a The movement of people into a country is known as:
A push factors
B pull factors
C immigration
D emigration
 - b In 2002–03 the largest number of migrants to Australia was from
A China
B India
C New Zealand
D United Kingdom
 - c Between 1982–83 and 2002–03 the country which experienced the largest percentage increase in migrants to Australia was
A China
B India
C New Zealand
D South Africa
 - d The category of migrants which made up the largest proportion of Australia’s migration program in 2001–02 was
A Family
B Skilled
C Humanitarian
D New Zealand
 - e The eligibility category which experienced the largest percentage change in migrants to Australia between 1996–97 and 2001–02 was
A Family
B Skilled
C Humanitarian
D New Zealand

20

REGIONAL AND GLOBAL LINKS – TRADE

Introduction Trade is an essential regional and global link for Australia. It allows us to export products in which we have a competitive advantage and import products which we are less efficient at producing. The Australian government pursues a multilateral (through its membership of the World Trade Organisation), regional (through its membership of regional organisations such as APEC) and bilateral (through negotiation of specific free trade agreements) approach to trade policy.

Benefits of trade

1 Make a list of the benefits to Australia of trade.

Terminology

2 Match each of the following terms with the correct definition below.

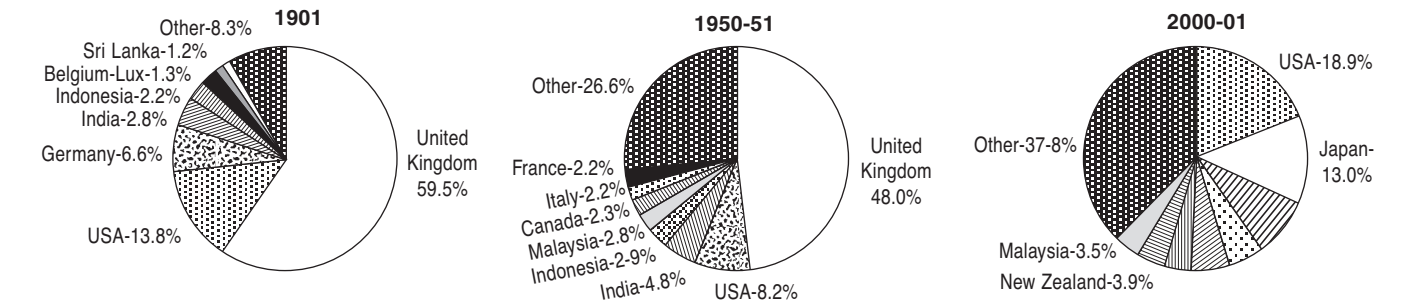
APEC, CERTA, Doha Round, ETMs, free trade agreement, SAFTA, tariff, World Trade Organization

a	A group of 21 Asia-Pacific nations that have an informal agreement to achieve free trade among its members by 2010 for developed countries and 2020 for developing countries.	
b	An agreement by which parties agree to eliminate tariffs and other trade barriers between themselves.	
c	Elaborately transformed manufactures. They include machinery, motor vehicles, computer equipment and telecommunications equipment.	
d	A tax on goods which have been produced overseas, imposed by the government of the country to which they are exported	
e	This round of trade talks were launched in 2001. It has been of particular concern to Australia because of its focus on reductions for agricultural protection.	
f	The bilateral free trade agreement which has been negotiated between Australia and Singapore.	
g	This bilateral agreement has led to the removal of trade restrictions between Australia and New Zealand.	
h	A global organisation, dealing with the rules of trade between nations. It was established in 1995 to replace GATT. It has 148 member nations, accounting for 97% of world trade. The objective is to 'help trade flow smoothly, freely, fairly and predictably'.	

Changing direction of Australia's trade

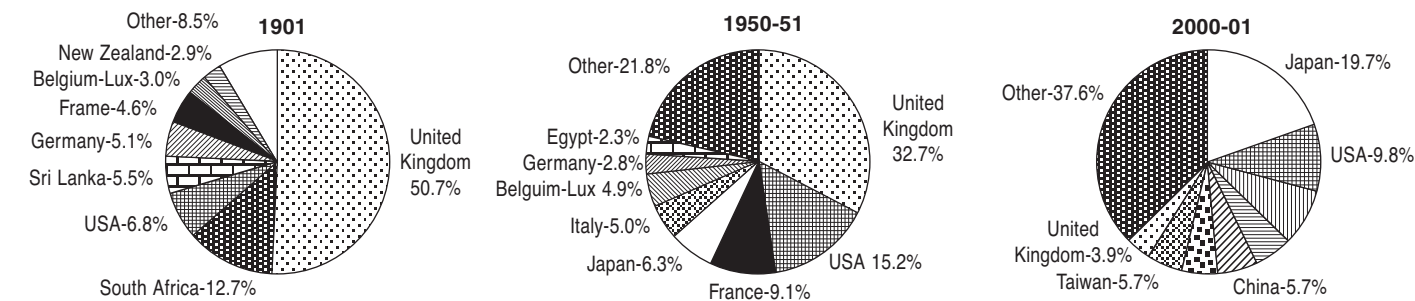
- 3 Refer to the pie graphs on this page and the other set over.
- a Using statistics compare the direction of Australia's trade in 1901 and 2000–01.
- b Suggest reasons for the changes in the direction of Australia's trade.

Australia's merchandise imports by top eight countries



[Source: 'Direction of Trade Time Series', Department of Foreign Affairs and Trade, February 2002]

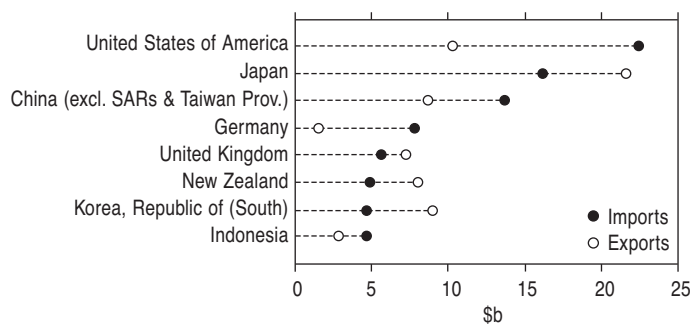
Australia's merchandise exports by top eight countries



[Source: 'Direction of Trade Time Series', Department of Foreign Affairs and Trade, February 2002]

Australia's trading partners

Merchandise imports and exports, selected countries, 2002-03



[Source: Australian Bureau of Statistics, Yearbook Australia, 2004]

4 Explain each of the following terms:

- trade deficit
- trade surplus

5 From the graph above make a list of the countries with which Australia had a trade surplus in 2002-03 and those with which Australia had a trade deficit.

Composition of Australia's trade

Major Australian Exports, 2003-04	(A\$m)
Coal	10 893
Non-monetary gold	5 651
Iron ore	5 216
Crude petroleum	4 643
Bovine meat	3 917
Major Australian Imports, 2003-04	(A\$m)
Passenger motor vehicles	11 217
Crude petroleum	6 321
Computers	5 127
Medicaments (incl. veterinary)	4 898
Telecommunications equipment	4 359

True or false?

6 Use your own knowledge and the stimulus material above. Label the statements a-e in the right column True or False. Rewrite the incorrect statements correctly.

- a Australia is a major importer of primary products and a major exporter of ETMs.
- b Australia's imports indicate that Australian consumers are vulnerable to fluctuations in the world oil price.
- c The World Trade Organisation is an example of a bilateral trade agreement.
- d Australia tends to import primary products rather produce them because high labour costs in Australia means it has difficulty competing with products produced overseas.
- e The Australian government wants to see a reduction in agricultural protection at the global scale.

Multiple choice

7 Circle the answer that best suits the question.

a In 1901 _____ % of Australia's merchandise imports came from the UK

- A 51
- B 181
- C 215
- D 59.5

b In 2000-01 the nation that received the largest share of Australia's exports was:

- A USA
- B China
- C Japan
- D other

c Australia has its largest trade surplus with

- A USA
- B Japan
- C Germany
- D Republic of Korea

d The process by which trade barriers have been reduced is best described as

- A The World Trade Organization
- B bilateral free trade agreements
- C trade liberalisation
- D multilateralism

e Approximately three-quarters of the world's nations belong to

- A CERTA
- B bilateral trade agreements
- C SAFTA
- D The World Trade Organization

SELECTED ANSWERS

1 INTRODUCING AUSTRALIA'S PHYSICAL GEOGRAPHY

- 1a** (1) 10° S (2) 44° S (3) 113° E (4) 154° E (5) 3983 (6) 3138
- 2a** (1) largest (2) 7682 300 (3) flattest (4) 2228 (5) driest
- 2b** (1) Cape York (2) South East Cape (3) Cape Byron (4) Steep Point (5) Mt Kosciuszko (6) Murray–Darling
- 3a** It is warm all year round in Darwin because of its close proximity to the equator where the sun is directly overhead. Between December and March Darwin experiences heavy rainfall due the arrival of the monsoon. **3b** Temperatures fall with altitude – approximately 6°C for every 1000 metres. **3c** Alice Springs is a long way from the coast so therefore does not get any rain bearing coastal winds. **3d** Located at 33°S, Sydney is much closer to the equator than Hobart at 42°S. Sydney therefore receives more direct sunlight. **3e** The sub-tropical high migrates south in summer and dominates this region, blocking the intrusion of maritime air.
- 4a** (1) Western (2) Central (3) Eastern
- 4b** fold mountains/Eastern Highlands, block mountains/Flinders Ranges, volcano/Mount Gambier, dissected plateau/Kimberley, granite boulders/Devils Marbles, limestone plain/Nullarbor, sandstone monolith/Uluru, sedimentary plateau/Carnarvon Gorge
- 5** **a** Ningaloo Reef **b** Kangaroo Island **c** Fraser Island **d** Macdonnell Ranges **e** Wave Rock **f** Lake Mungo **g** Daintree Rainforest **h** Cradle Mountain
- 6** **a** False. Cape York Peninsula is north-east of Alice Springs. **b** True **c** False. The Flinders Ranges are located in South Australia. **d** True **e** False. It is much hotter in the northern parts of Australia than in the south. **f** False. Perth's climate can be described as Mediterranean.

2 INTRODUCING AUSTRALIA'S HUMAN GEOGRAPHY

- 1a** 1 Rockhampton 2 Perth 3 Alice Springs 4 Mount Isa 5 Geelong 6 Adelaide 7 Mackay 8 Hobart 9 Broken Hill 10 Albany
- 1b** Abbreviations: ACT, NSW, NT, Qld, SA, Tas, Vic, WA ACT places – Belconnen, Tuggeranong, Weston Creek; NSW places – Broken Hill, Coonabarabran, Tamworth; NT places – Alice Springs, Darwin, Katherine; Qld place – Cairns, Gold Coast, Longreach, Weipa; SA places – Coober Pedy, Mount Gambier, Whyalla; Tas places – Devonport, Launceston, Queenstown; Vic places – Bairnsdale, Portland, Shepparton; WA places – Albany, Broome, Kalgoorlie,

City	Population '000	Location
Sydney	4 085.6	Located on Australia's eastern seaboard at 33.55S, 151.13E
Melbourne	3 466.0	Located at 37.49S, 144.58E on Port Phillip Bay on Australia's south-eastern seaboard
Brisbane	1 626.9	Located at 27.28S, 153.01E on Moreton Bay on Australia's eastern seaboard
Adelaide	1 381.1	Located at 34.36S, 138.36E on the eastern shore of the Gulf of St Vincent on Australia's southern coast
Perth	1 096.1	Located at 31.57S, 115.51E on the Swan River on Australia's south-west coast
Canberra	310.5	Located approximately midway between Sydney and Melbourne in the Eastern Highlands at 35.17S, 149.08E
Hobart	194.2	Located at 42.50S, 147.15E on the south-eastern side of Tasmania at the mouth of the Derwent
Darwin	90.0	Australia's northernmost city on the Van Dieman Gulf at 12.27S, 130.50E

- 3a** manufacturing, retail trade, health and community services
3b retail trade, manufacturing, property and business services
- 4** **a** wool industry **b** Newcastle **c** Weipa **d** manufacturing
e tourist industry **f** wine industry

5 D **6** A **7** A **8** B **9** C

3 LANDFORM

- 1** weathering – gradation, folding – tectonic, faulting – tectonic, erosion – gradation, volcanism – tectonic, landslide – gradation, deposition – gradation
- 2** **a** valley – erosional, **b** delta – depositional, **c** mesa – residual, **d** canyon – erosional, **e** gibber plain – erosional/residual, **f** ridge – residual, **g** alluvial fan – depositional, **h** plain – residual/depositional, **i** cirque – erosional, **j** beach – depositional, **k** mountain – residual, **l** dune – depositional, **m** plateau – residual, **n** crater – residual
- 3b** 837 metres – as the continents are different sizes, their size would affect the real world average/calculating an average without including size gives a less accurate result
- 4** B, **5** C, **6** A, **7** C, **8** B

4 WATER RESOURCES AND DRAINAGE

- 1** In order 1–10: 1. irrigation 2. aquifer 3. gigalitre 4. perennial 5. external 6. artesian 7. internal 8. catchment 9. discharge 10. intermittent
- 3** Graph sector sizes: rivers 263°, groundwater 75.5°, other surface water 21.5°
- 4** **a** population and water use follow the same general trend **b** NSW and Qld **c** ACT – (little irrigated agriculture or heavy industry to use water and high population to land ratio) **d** More people, more water used, NSW, Qld – large states and rural areas to irrigate, considerable industry. Vic – large population, much industry, much irrigation from Murray and tributaries SA and WA have large arid areas/less water available for extensive irrigation, fewer people, less industry Tas – smaller areas and population, humid climate so less irrigation required than on mainland **e** NSW 10%, Vic 10%, Qld 35%, WA 63%, SA 36%, Tas 4%, NT 71%, ACT 7% **f** Groundwater use depends on both need and availability. Those states with more arid climates or extensive arid regions are more likely to use it – mainly for stock watering and irrigation, so farmland quality is also a factor – SA has less available than other mainland states. Tas has less need with more surface water. Qld and WA have the largest artesian basin areas

5 broken lines intermittent, solid lines perennial – intermittent rivers in Lake Eyre basin are internal drainage

6 B, **7** D, **8** A, **9** B, **10** C

5 AIR MASSES AND CLIMATE

- 1** E – equatorial, hot and humid, northern areas most of the year; Tm – tropical maritime, warm and humid, northern areas in winter and further south in summer; Tc – tropical continental, warm and dry, central Australia all year; Sm – southern maritime, cool and humid, southern areas most of the year, extending further north in winter; Pm – polar maritime, cold and humid, southern areas mainly in winter; Pc – polar continental, cold and dry, does not influence Australia
- 2** 1 D Tropical Wet–Dry, Monsoon 2 C Tropical East Coast 3 B Tropical Desert 4 A Mid Latitude Continental 5 G Warm Temperate West Coast, Mediterranean 6 F Warm Temperate East Coast 7 E Cool Temperate West Coast

6 WEATHER PATTERNS

- 1** temperature, humidity, precipitation (amount, type, duration), cloud

cover (type, extent), wind (speed and direction), air pressure

2 clothing, food and beverage consumption, recreational activity, housing (type, maintenance, energy consumption) transport, human energy/activity levels, psychology (mood), employment patterns (industry)

3 B, 4 C, 5 C, 6 A, 7 D

8 Adelaide – cold, cloudy, raining, strong wind 33-41kph from SW, high humidity, pressure 1013 hPa; Darwin – warm, clear to light cloud, fine (no rain) moderate wind 24-32 kph from ESE, pressure 1014 hPa

9 Pressure patterns move from west to east. Winds will initially continue from SW – SSW, then ease. As the High moves over Adelaide, clearing skies, rain easing– fine cool weather will prevail. As the high moves to the east of Adelaide, winds will become WNW, clear dry condition will continue and temperatures will be a little warmer.

11 B, 12 C, 13 C, 14 D, 15 C

16 Summer brings hot humid conditions to the North Queensland coast. NW winds would bring hot humid tropical maritime air into Townsville, making conditions hot and humid with considerable cloud cover. This air may rise in convectional thunderstorms as it crosses onto and is heated by the land surface, and orographic precipitation would occur as the air passes over the Great Dividing Range to the west of Townsville.

17 Sydney is experiencing very hot and dry weather, clear skies and low humidity due to a strong NW wind bringing Tropical Continental air from the NSW, QLD, NT border region. The cold front is moving to the east, and within 24 hrs will pass through Sydney. A change of wind direction to the SW will lower temperatures due to a Southern Maritime air mass, humidity will increase and winds ease. Cool, but not cold weather with some cloud and the possibility of light rain can be expected. Heavy rain and thunderstorms are unlikely as the air forced up by the cold front was very dry.

7 VEGETATION AND FOREST RESOURCES

1 B, D, E, A, C

2 Sector sizes: Open forest 17°, Woodland 143°, Shrubland 134°, Grassland 35°, Pastures 23°, Other 5°, Coastal and Rainforest approx. 1.5° each.

3 F, T, T, F, T

4 Utility – firewood, timber for building/construction, furniture, paper products, packaging, honey, flowers, bark and woodchips, resins (from sap), shade, eucalyptus oil, medicinal products, bushwalking/camping, power poles etc; Environmental – reduce soil erosion, improve air quality/greenhouse gas (carbon sink, oxygen source), animal habitat (birds, koala possum, insects etc), reduce water runoff, improve soil fertility (humus), plant habitat (epiphytes), reduce soil evaporation, animal food source (koala, insects etc); Economic– timbergetters, sawmillers, truck drivers, forest research workers, national park staff; Aesthetic/intrinsic – appearance, beauty/attraction, shade, biodiversity maintenance, artistic subject

5a Forests – 38%, Woodlands 19%. Forest timber is more valuable, useful than woodland, there are much larger initial areas of woodland, settlement and landuse are more sparse in woodland zones.

5b Direct use – timber, woodchips, fuelwood (eg. domestic, industrial transport– Murray River Paddle Steamers) and clearing for alternative uses – eg. agriculture, transport routes, mining, dams and reservoirs, industrial land, urban expansion

8 SOILS AND SALINITY

1 1 lithosphere, 2 weathered, 3 organic, 4 reactions, 5 horizons, 6 profile, 7 soluble, 8 topsoil, 9 infertile, 10 replenish, 11 salinity,

12 leached, 13 volcanic, 14 alluvial, 15 intensive.

2 C, 3 C, 4 B, 5 D, 6 C, 7 D

9 Areas not shown would either be not significantly affected by salinity eg NSW coastline and/or not used intensively for farming eg. Central Australia forested and urban land– WA is most, and Qld. and the NT least. Much of the WA problem is dryland salinity, with other mainland states having areas of both dryland and irrigation problems. Irrigation is significant in the Murray, Murrumbidgee and Ord valleys and SE Qld. Salts dissolved from the upper reaches of river systems eg, SE Qld are carried and accumulate in the lower reaches eg. SE of South Australia.

10a Sector sizes in order – 1 excess to allocated from rounding – 18°,13°, 9°, 8°, 7°, 6°, 4°, 9°, 99°, 86°, 84°, 16°

10b Non irrigated farming affects far more farms, with grazing and crop production being most affected, suggesting that dryland salinity is more widespread than irrigation salinity. Due to different numbers of each farm type and the areas they cover, other measures would be needed to examine the impact further, but crop and livestock farming usually covers a greater area of land than more intensive farming types.

9 NATURAL HAZARDS

1a Lithosphere – the solid inorganic outer surface of the earth’s crust (composed of rocks minerals and elements); Atmosphere – the vast gaseous envelope of air that surrounds the earth whose boundaries are not easily defined. It contains a complex system of gases and suspended particles; Biosphere – the zone of the earth where life is found so consists of all living things, plant and animal; Hydrosphere – the waters of the earth in various stores, including oceans, lakes, rivers, glaciers, snowfields, groundwater and particles of water vapour in the atmosphere.

1b Lithosphere – Geologic, earthquakes, volcanoes, tsunamis – Geomorphic, landslides, rockfall, earthflow, mudflow, dune migration, dust storm; Atmosphere – severe storms (wind, hail, lightning), cyclones, tornadoes, heatwaves, drought, bushfire, dust storm; Hydrosphere – flood, tsunamis, storm surge; Biosphere – bushfire, pests/plague, disease (bacterial, viral). Note: some hazards may be classified in more than one category, eg. extra terrestrial hazards like meteorites could be lithosphere, atmosphere or a separate category.

2 Lowest in eastern inland and Nullarbor Plain sparse (low fire risk, low population density, main risks from flooding in east and earthquakes (low), Highest around SE coast. SW corner and along NW coast. Risk higher in SE/SW due to population density, forest fires and local flooding – potential for storm and earthquakes. NW prone to cyclones, flooding and earthquakes – less chance of earthquakes on NE coast. Main factors are the nature of the environment and the population density.

3 a Meckering Earthquake	b Ash Wednesday Bushfires
c Thredbo Landslide	d Great Influenza epidemic
e Newcastle Earthquake	f Hobart Bushfires
g Cyclone Tracy	h Southern Australian Heatwave
i Sydney Hailstorm	

4 Sector sizes – 119°, 79°, 68°, 32°, 22°, 22°, 14°, 4°

5 T, T, F, F, F, F

10 DROUGHTS AND BUSHFIRES

1a In high pressure systems, air descends and becomes warmer. The increased heat reduces the relative humidity, making the air capable of evaporating moisture. El Nino occurs when winds that normally blow warm, easily evaporated water toward Australia across the Pacific fail, and our region is dominated by colder surface waters and higher air pressure. The warm waters normally combine with lower pressure to bring rain. 1b Our small area of significant highland limits orographic (relief) rainfall, while cold ocean currents

on the west coast contribute to aridity there. Pronounced dry seasons are experienced due to migration of high pressure belts in the north and south west of the continent.

2b Most severe in the east, with NSW most affected, then Victoria and Queensland. South Australia is also significantly affected. Small areas of WA and NT have severe drought, but they also have areas of average and pockets of above average conditions. Tasmania has average conditions in the north, but high rainfall in the south west and drought in the east.

4b Response – handfeeding, agistment, reducing spending, off-farm employment, improving irrigation efficiency, destocking; Preparation – storing grain, building water storages, storing fodder. Some responses could be considered preparation if undertaken very early in a drought cycle

4c Handfeeding – expensive but allows farmers to retain breeding stock; Agistment – allows stock to be retained and returned if conditions improve; Reduced spending – reaction to reduced income/helps keep farm viable

off-farm employment – as for reduced spending; Irrigation efficiency – increase potential production, educes costs; Destocking – educes demand for scarce/expensive feed; Storing grain – can be used for stock feed in future drought

water storages– enables production to continue at higher level/ longer in droughts; Storing fodder – as for storing grain

5 No. of bushfires – 5995; Area burnt – 21241 ha; Fatalities 7; Houses destroyed 646; Stock lost 14 427

6 C, 7 D, 8 D, 9 D, 10 B

11a Causes in table order – 26%, 25%, 16%, 10%, 7%, 5%, 3%, 2%, 1% Areas burnt in table order – 46%, 14%, 7%, 1%, <1%, 3%, 9%, 2%, 5%, 14%

11b Causes from table, impacts from previous table plus expanded material eg. loss of personal effects, emotional impacts, economic impacts, review of reactions regulations and practices. Droughts dry out fire fuels, high temperatures add to ease of ignition, winds supply more oxygen and cause fire fronts to move faster, low humidity/easier ignition and spread, lack of precipitation dries and adds to fuels.

Fires on town/urban fringes most significant for danger to property and human life – significant stock and crop losses can occur in more intensively used rural zones – less danger from fires in natural bushland or extensively farmed areas (eg. prolific lightning lit dry season scrub and grass fires left to burn in the NT)

11 DEMOGRAPHIC CHARACTERISTICS

2	% share
NSW	33.7
VIC	24.7
QLD	18.8
SA	7.7
WA	9.8
TAS	2.4
NT	1.0
ACT	1.6
AUST	100.0

3a Australia’s overall population size has increased. Numbers of both males and females in the age groups below 34 have declined. Numbers of both males and females in the age groups above 35 have all increased.

3b Women are having fewer children because they are pursuing a

career, leaving having children until later in life. Larger numbers of women are choosing to have no children at all. People are now living longer due to better health, nutrition, living standards and improved medical services.

5 The trend in Australia is an increase in the number of people in the 65 and over age group. This ageing group will represent a much larger proportion of the total population than ever before. At the same time, the proportion of younger

Australians is decreasing. There will be fewer working age people relative to non-working age people.

7 a True **b** False. In 1983 the largest group of Australians was the 10–14 year age group by 2003, the largest group was the 40–44 year age group. **c** False. In 1901 23% of Australia’s population was overseas born. **d** True **e** False. The Italian, Greek and Dutch populations which all experienced an increase after WWII due to post-war migration have all declined in recent years.

f True

8 C, 9 B, 10 D, 11 B, 12 B

12 POPULATION DISTRIBUTION

1 Dense around SE coast from Brisbane to Adelaide, around Perth – moderate in most of Tasmania and Qld coast below Cairns, remainder sparse

2 History – settlement took place from the outside, so our population was focussed on a small number of initial locations from the beginning Climate is more pleasant in the southern regions (high temps, humidity and cyclones in north), most even rainfall in east, dry inland with temperature extremes; Transport Routes – radiated from initial city settlements reinforcing the pattern, and developed along the coastlines between them; Economic Factors – most employment has always been in the established cities, since strengthened by farm mechanisation and rural decline; Water Supplies – more abundant and reliable on the east and north coast, less in west and centre; Environmental Amenity – 85% of Australians live within 50 km of the coastline, which brings climatic and other benefits/densely settled areas generally have higher natural amenity and better access to services

3 Sector sizes – NSW 122°, Vic 89°, Qld. 67°, SA 28°, WA 35°, Tas. 9°, NT 4°, ACT 6°

4 D, 5 B, 6 D, 7 B, 8 D, 9 A

10 Reasons – sparse rural landuse (grazing), mining settlements, isolated tourist attractions, aboriginal settlements (esp. NT); Difficulties- (vary with settlement type) low incomes, high unemployment, limited job range, sexual imbalance, age imbalance, limited goods and services, limited transport, long travel times to major centres, limited social interaction

11 Determined by categories selected

12a Gold Coast –Tweed, Mandurah, Sunshine Coast, Bunbury, Townsville **b** All coastal, most close to but not major cities **c** ageing population /retirement, major city overflow from city problems/prices, employment opportunities, amenity and lifestyle opportunities **d** rising housing prices, strain on services, increasing traffic congestion

13 Advantages – access to widest range of goods, services and employment; Problems – high housing costs, traffic congestion, travel times, pollution, crime, ageing infrastructure, stress on services eg. public transport, water supplies; Solutions – decentralisation, urban villages, investment in public transport and infrastructure

13 INDIGENOUS AUSTRALIANS

2 b 1981 and 1986 and 1996 and 2001 **c** high fertility rate and more people recognising themselves as indigenous.

3 a Indigenous Australians – 2.8%; non-indigenous Australians –

13% **b** Poor health, lack of access to medical facilities and low levels of education have led to fewer indigenous people living over 65
c Indigenous Australians– 39%; Non-Indigenous Australians – 20.4%
d The fertility rate of indigenous women is higher than the fertility rate of non-indigenous women.

4 a Non-indigenous males have a life expectancy 21 years greater than indigenous males; Non-indigenous females have a life expectancy 19 years greater than indigenous females **b** The huge disparity in life expectancy can be attributed to poor health and nutrition, low levels of education and poor access to medical services of indigenous people.

5 a True **b** False. Indigenous Australians have a younger population structure than non-indigenous Australians.
c False. The largest group of indigenous Australians in 2001 was the 5-9 year age group. **d** False. The life expectancy of non-indigenous females is 19 years greater than indigenous females.
e True

14 NATIVE TITLE

1 a mediation; **b** Mabo; **c** coexistence; **d** Crown land; **e** native title; **f** freehold land; **g** Native Title Tribunal; **h** pastoral lease; **i** terra nullius; **j** Wik

		Event	Impact
3 June 1992		High Court's Mabo Decision	The doctrine of terra nullius was overturned. For the first time it was recognised that indigenous Australians may continue to hold native title over their land.
1993		Creation of the Native Title Act	The Act established the Native Title Tribunal
1994		Establishment of the Native Title Tribunal	A more formal process was created by which indigenous Australians could resolve native title applications.
December 1996		High Court's Wik Judgement	It was recognised that pastoral leases may not extinguish native title.

	Possibly	Definitely not	Explanation
A		✓	Schools extinguish native title.
B	✓		Native title can exist on some types of pastoral leases.
C		✓	Privately owned land extinguishes native title.
D	✓		Native title is not extinguished on vacant crown land.
E		✓	A commercial lease extinguishes native title.

4 a False. Native Title was first recognised in Australia in 1992 when the High Court recognised that the Meriam people of Torres Strait had native title over their lands. **b** True **c** False. Determinations about Native Title are made by the Federal Court.
d True **e** True

5 1 terra nullius; 2 1992; 3 Mabo; 4 Native Title Act; 5 Native Title; 6 mediate; 7 Federal Court of Australia; 8 54; 9 35

15 OUR CHANGING AGE STRUCTURE

	Fertility rate	Possible reasons
1921	3.2	Good economic times
1934	2.1	The Great Depression
1948	3.1	Birth rate increased following WWII
1966	2.9	Oral contraceptive pill available
1979	1.9	Increasing participation of women in education and the labour force, greater access to contraception and abortion
2001	1.7	Women leaving child birth to later in life and therefore having fewer children

2 a males – 25; females – 26 **b** better health, better nutrition, better access to medical services

	Change
Total population	15 889
Proportion of the population aged (years)	
0–14	14.8
15–64	6.3
65–84	7.3
85+	1.3
Median age	13.4

3b Australia's population is ageing: there has been a decline in the percentage of people in the 0-14 age group; there has been an increase in the percentage of the population in the 14-64 year age group, the 65–84 year age group and the 85 and over

4 B, 5 A, 6 A, 7 D, 8 A

16 INDUSTRIAL LANDSCAPE

1 primary 8% – agriculture, fishing and forestry (3%) and mining (5%); secondary – manufacturing 11%; tertiary – all others 81%

2 a D, b B, c D, d C, e C, f B

3a Sector sizes – NSW and Vic 115°, Qld 54°, WA 32°, SA 25°, Tas 11°, NT and ACT 4°

4 a Kwinana; **b** Mt. Isa; **c** Adelaide; **d** Mackay, **e** Canberra; **f** Newcastle; **g** Gladstone; **h** Toowoomba; **i** Brisbane; **j** Melbourne; **k** Geelong; **l** Dubbo, **m** Burnie; **n** Wollongong–Port Kembla; **o** Shepparton

17 REGIONAL AND GLOBAL LINKS – DEFENCE

Australia's Defence Force Operations– 2004

1b Australia's defence force operations are predominantly located within the Asia Pacific region, the Middle East and Africa

1c 1 E; 2 J; 3 G; 4 D; 5 F; 6 I; 7 C; 8 B; 9 H; 10 A

2 a False. Defence spending has experienced an increase between 1993-94 and 2003-04. **b** True **c** True **d** False. The smallest area of the defence forces in terms of employment is the navy.

3 (1) Pacific (2) Australia (3) New Zealand (4) USA (5) San Francisco (6) 1951 (7) mutual (8) independent (9) interests (10) cornerstone (11) World Trade Centre (12) invoked (13) Article IV (14) terrorism

18 REGIONAL AND GLOBAL LINKS – INTERNATIONAL TOURISM

- 1b fear of terrorism, September 11, Bali bombing
- 2 a False. Between 1997 and 2002 total visitor numbers to Australia experienced a 12 percent increase. b True c True d False. People from the UK spent the largest number of visitor nights in Australia in 2002, comprising 19% of the total. e True
- 3 a More marketing and incentives for New Zealand tourists to come to Australia b Japanese economy slowing. c Asian economic crisis and improved conditions since d Growth of the Chinese economy and improvements in standard of living of the people e Marketing to tourist in the UK resulting in increasing numbers of back packers choosing to travel in Australia.

Region	Physical features	Human features
Sydney	Natural harbour and beaches	Taronga Zoo, Opera House, Harbour Bridge,
Melbourne	Fitzroy Gardens	Queen Victoria Market, Werribee Open Plains Zoo
Gold Coast	beaches of Coolangatta	Surfers Paradise, Warner Brothers Movie World
Tropical North	Great Barrier Reef, Daintree National Park	
Brisbane	Mount Coottha Lookout	South Bank
Perth	Kings Park,	Fremantle
Adelaide	Glenelg	Hahndorf
Petermann	Uluru, Kata Tjuta	
Whitsundays	Hamilton Island, Airlie Beach	
Sunshine Coast	Noosa National Park	Mooloolaba
Hervey Bay/	Fraser Island, whales	
Maryborough		
Northern Rivers	Mount Warning	
Tropical		
Alice Springs		John Flynn's grave, ANZAC Hill
Canberra		The War Memorial, The National Gallery,
Darwin	Litchfield National Park	Mindil Beach Markets

6 a C b D c A d D e C

19 REGIONAL AND GLOBAL LINKS – MIGRATION

- 1 1 net overseas migration 2 skilled migrants 3 pull factors
- 4 asylum seeker 5 emigration 6 family migrants 7 humanitarian migrants 8 immigration 9 push factors

	1982–83 %	2002–03 %
China	1.25	7.1
India	1.79	6.16
New Zealand	7.38	13.17
South Africa	2.97	4.90
United Kingdom	28.43	13.32
Vietnam	9.34	2.73

- 3
- | Push factors | Pull factors |
|-----------------------------|---------------------------|
| Political persecution | Employment |
| Unstable government | Better environment |
| War | Better standard of living |
| Poor economic circumstances | Stable government |
| Climate change | Education opportunities |
| Natural disaster | Socio-political tolerance |
- 4 (1) family (2) skilled (3) humanitarian (4) 88 900 (5) 3148 (6) Family (7) 42.5 (8) 23 344 (9) 26.3% (10) Skilled (11) 83% (12) 3154

- 6 a False. Australia's migration program has 3 eligibility categories. b False. During the period 1996-97 Australia experienced a shift in it migration focus from family migrants to skilled migrants. c False. In 2002-03 people from the United Kingdom comprised the largest proportion of migrants to Australia. d True

- 7 a C b D c A d B e B

20 REGIONAL AND GLOBAL LINKS – TRADE

- 1 Trade creates employment, creates income from exports, increases the range of good available in Australia and enables Australia to purchase products that cannot be produced as cheaply in Australia.
- 2 a APEC b free trade agreement c ETMs d tariff e Doha Round f SAFTA g CERTA h World Trade Organization
- 3a In 1901–50.7% of Australia's exports went to the UK but by 200–01 this had declined to 3.9% of Australia's exports. In 2000–01 the largest recipient of Australia's exports was Japan whose share of Australia's exports was 19.7%. In 1901 most of Australia's imports were from the UK (59.5%). In 2001–01 the largest share of Australia's imports came from the USA (18.9%).
- 3b In 1901 colonial links with the United Kingdom were still strong. They are now no longer as significant; the UK is a long way away and is more focused on trading with the European Union. Australia is now more focused on trading with countries in our region.
- 4 trade deficit – when imports are greater than exports; trade surplus – when exports are greater than imports
- 5 Trade deficit – USA, China, Germany, Indonesia; Trade surplus – Japan, UK, New Zealand, Republic of Korea
- 6 a False. Australia is a major exporter of primary products and major importer of ETMs. b True c False. The World Trade Organization is an example of a multilateral trade agreement. d False. Australia tends to import manufactured products rather than produce them because high labour costs in Australia means it has difficulty in competing with products produced overseas. e True
- 7 a D b C c B d C e D

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Extension work for advanced students	✓	A way of targeting student weaknesses	✓
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